

# 影像處理 02 數位影像基礎

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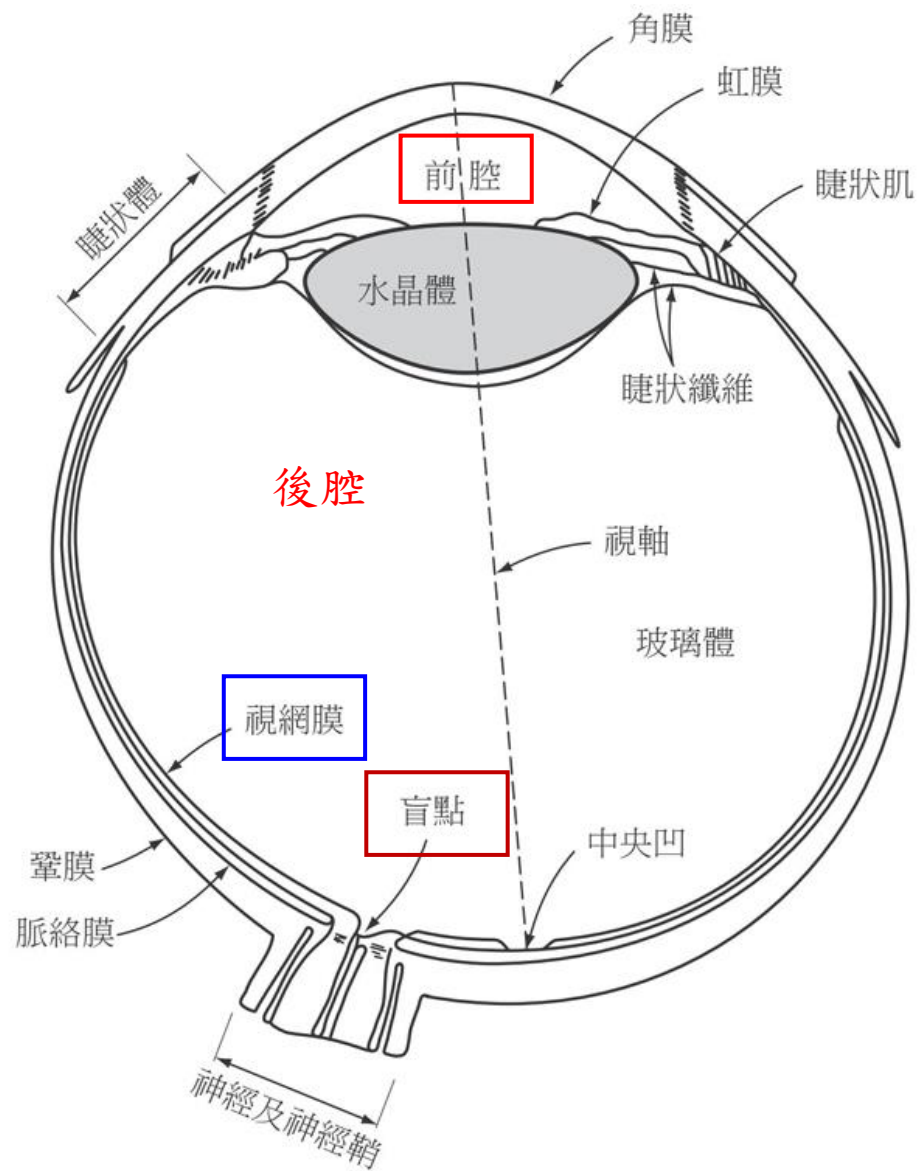
助教:陳信宏

# 大綱

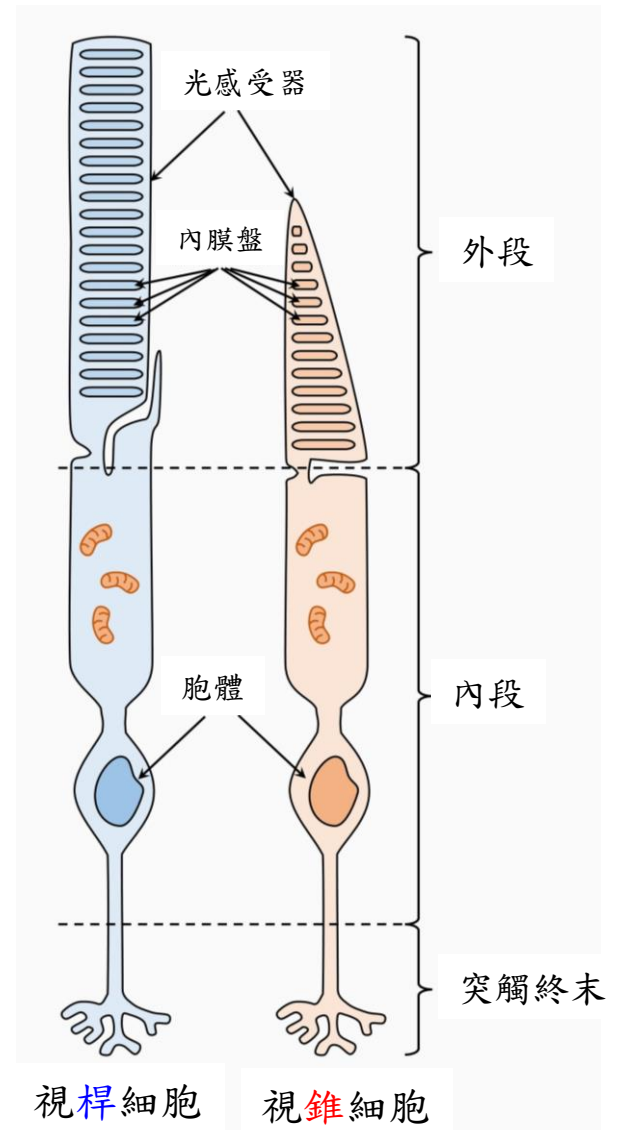
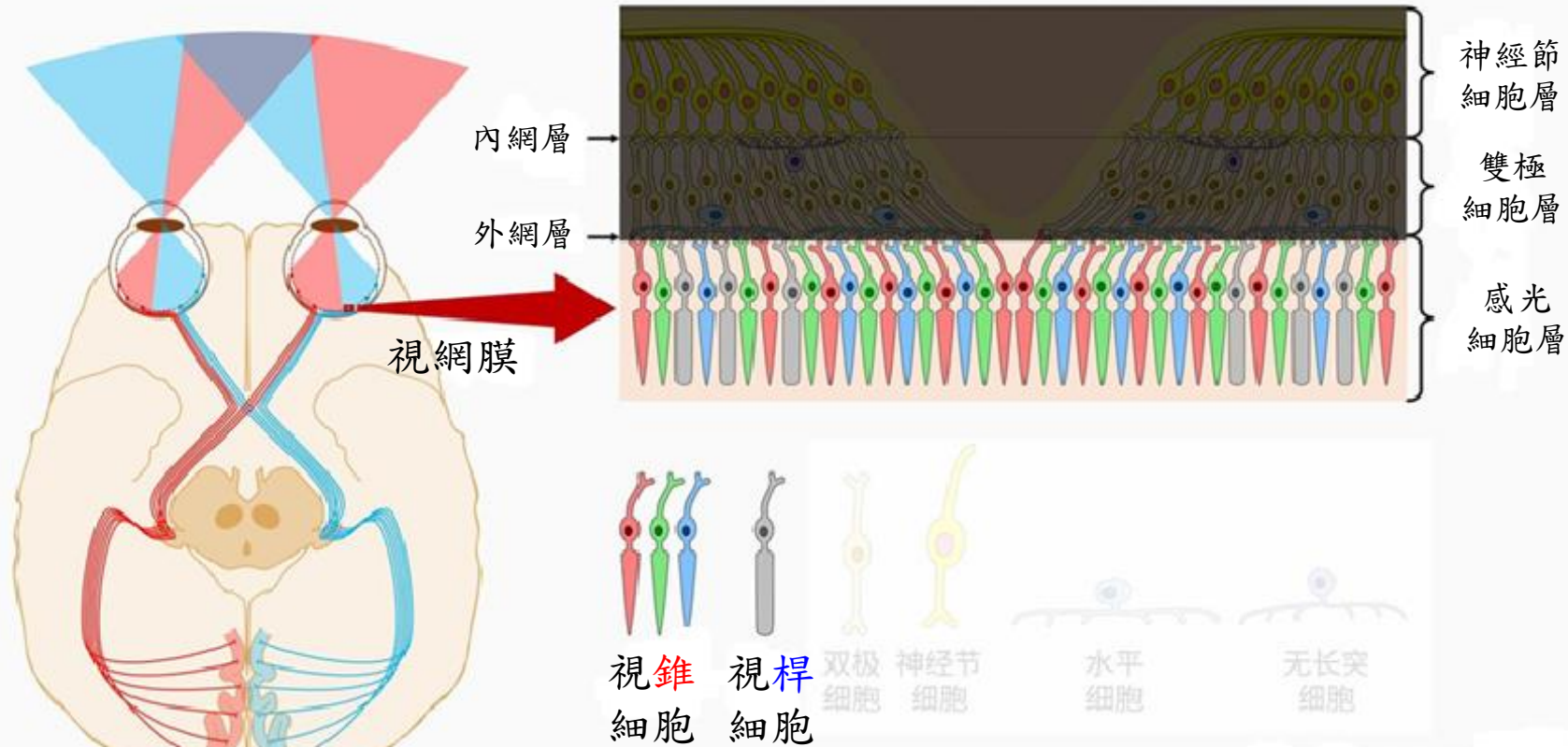
- 視覺感知
- 影像感應與擷取
- 影像取樣與量化
- 內插法(Interpolation)
- 空間轉換

## 2.1 視覺感知的要素

圖 2.1  
人類眼睛橫剖面的  
簡化圖形



# 桿細胞(Rod) & 錐細胞(Cone)



# 細胞分布

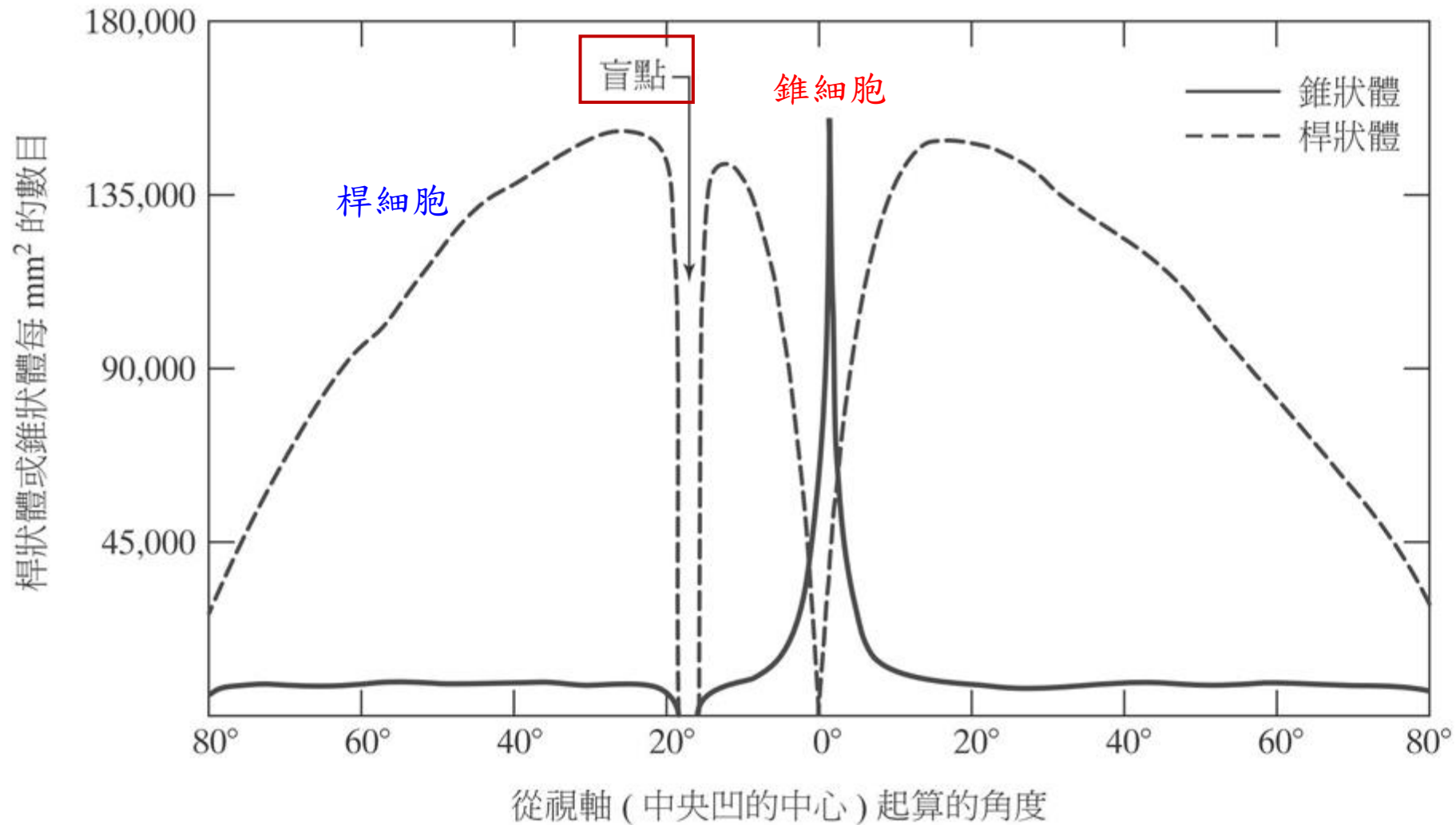


圖 2.2

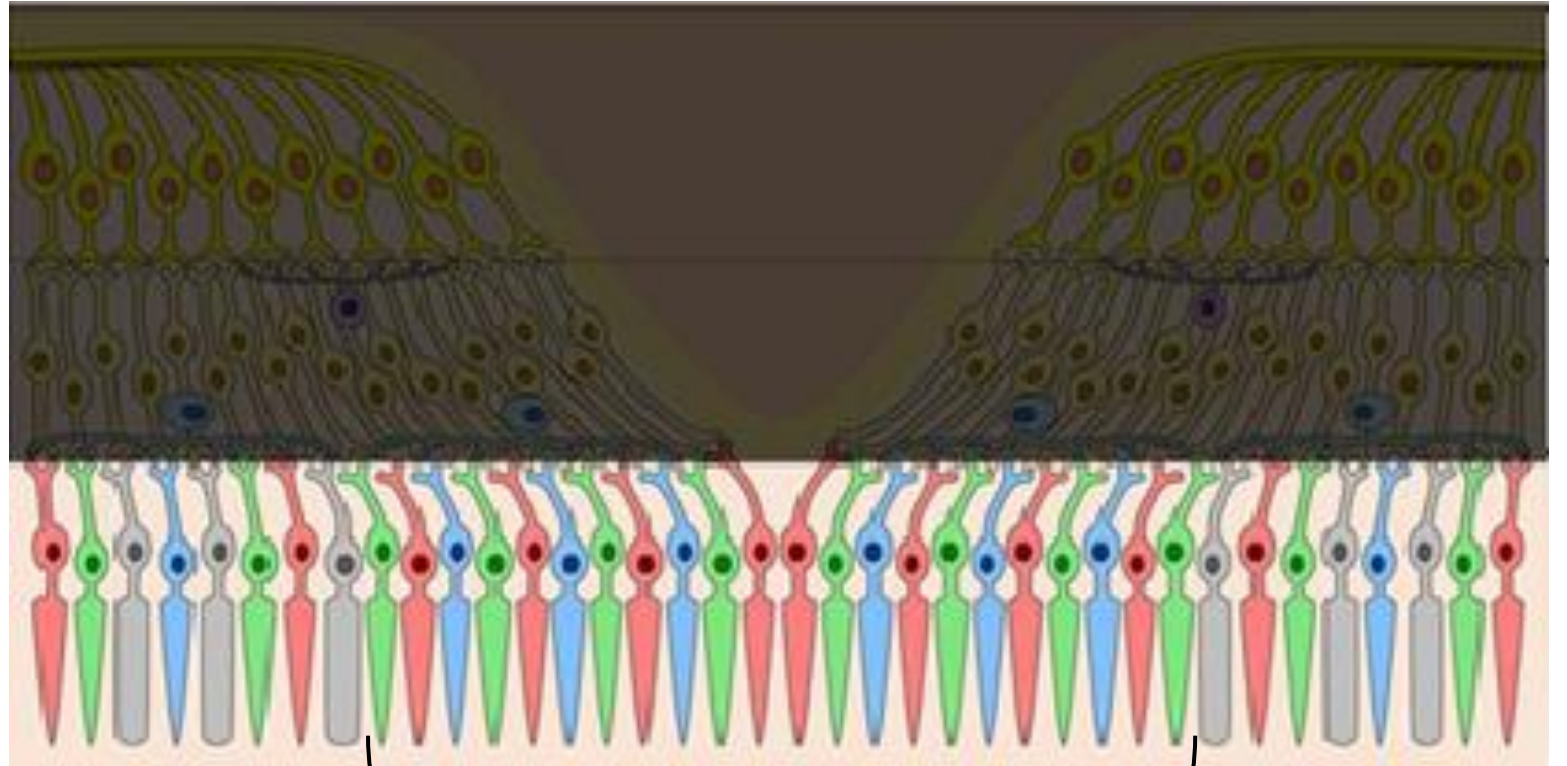
視網膜上桿狀體  
和錐狀體的分布

**錐細胞(Cone):**  
在亮光下工作  
負責顏色與細節辨識

**桿細胞(Rod):**  
在昏暗工作  
負責光強/區塊辨識



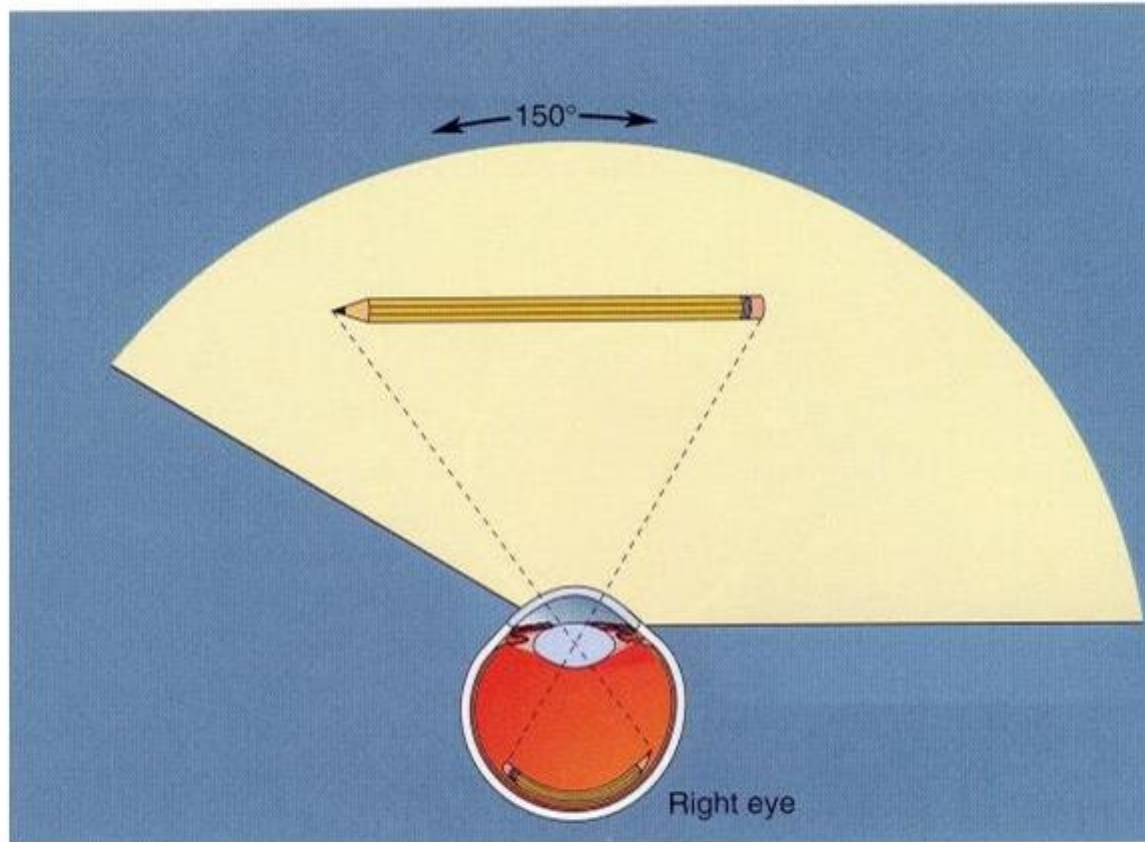
# 細胞分布



中間都是錐細胞

所以視覺聚焦看細節 都是眼睛要看過去

# The visual field

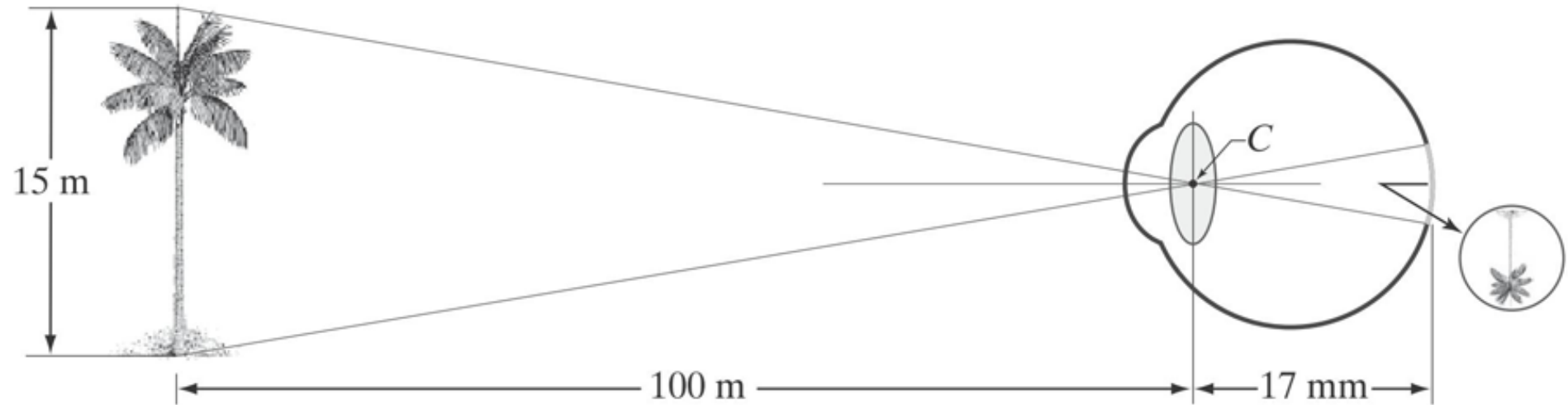


The image is **inverted**; the **left visual field** is imaged on the **right side of the retina**, and the right visual field on left side of the retina

# 眼睛中的細胞

圖 2.3

注視一棵棕櫚樹時眼睛的圖形表示。 $C$  點是水晶體的聚焦中心。



$$\frac{15}{100} = \frac{h}{17}$$

$$h = 2.25 \text{ mm}$$



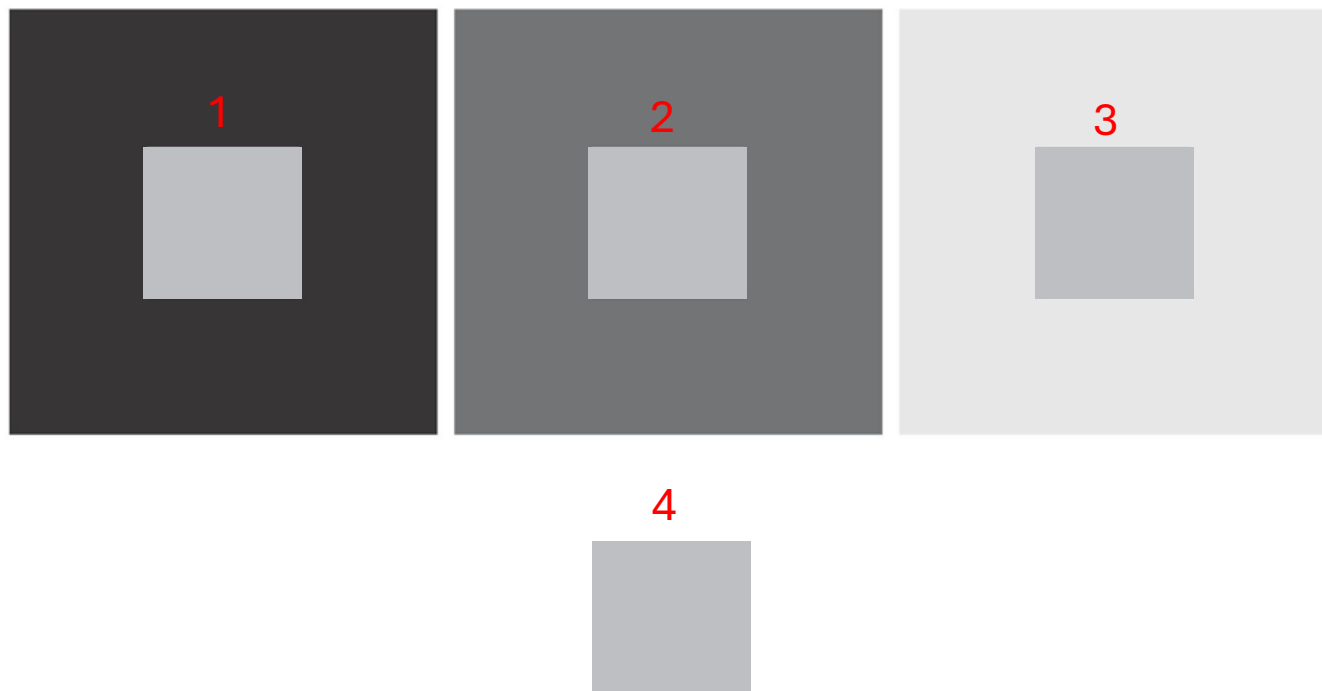
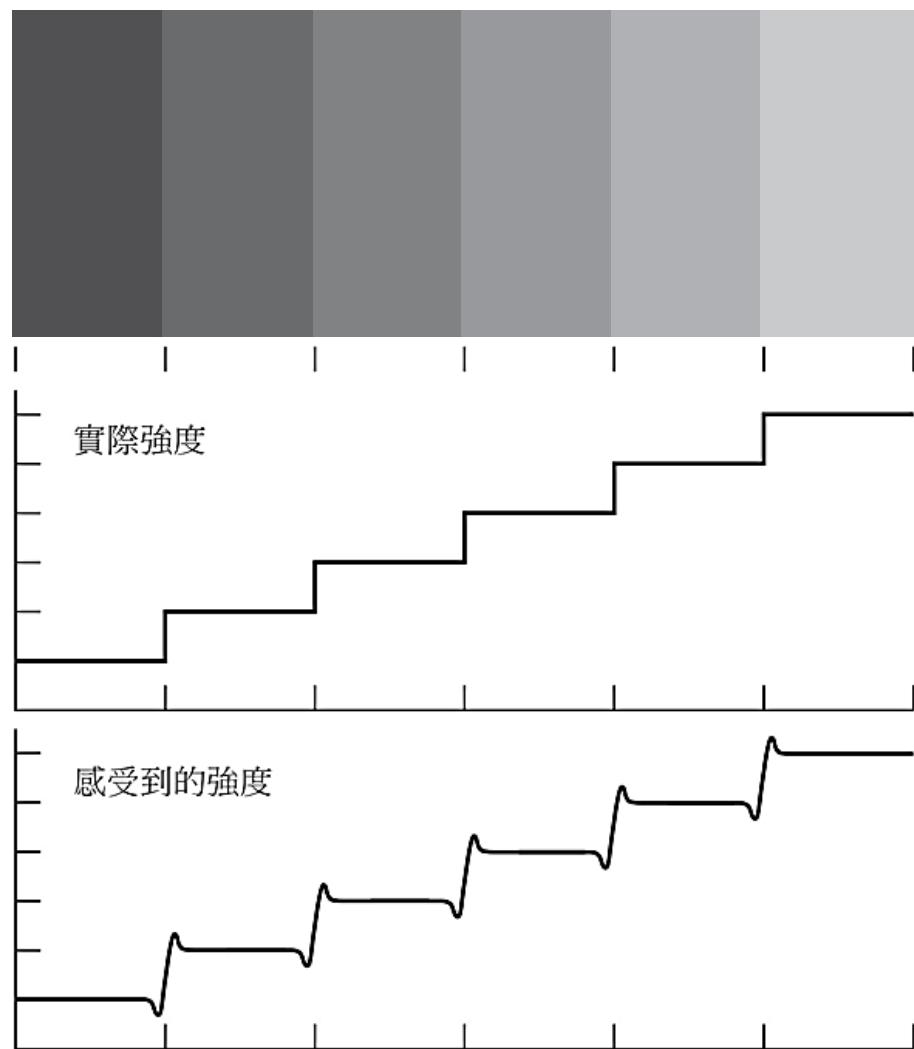
<https://www.youtube.com/watch?v=0HIYbNPCVts>

## 盲點實驗 | Find Your Visual Blind Spot

可能要近一點看

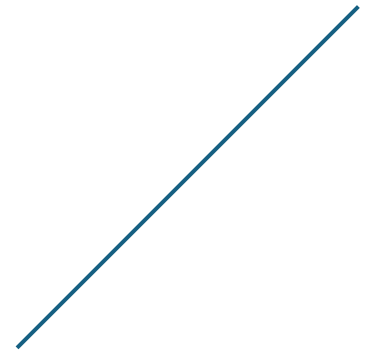
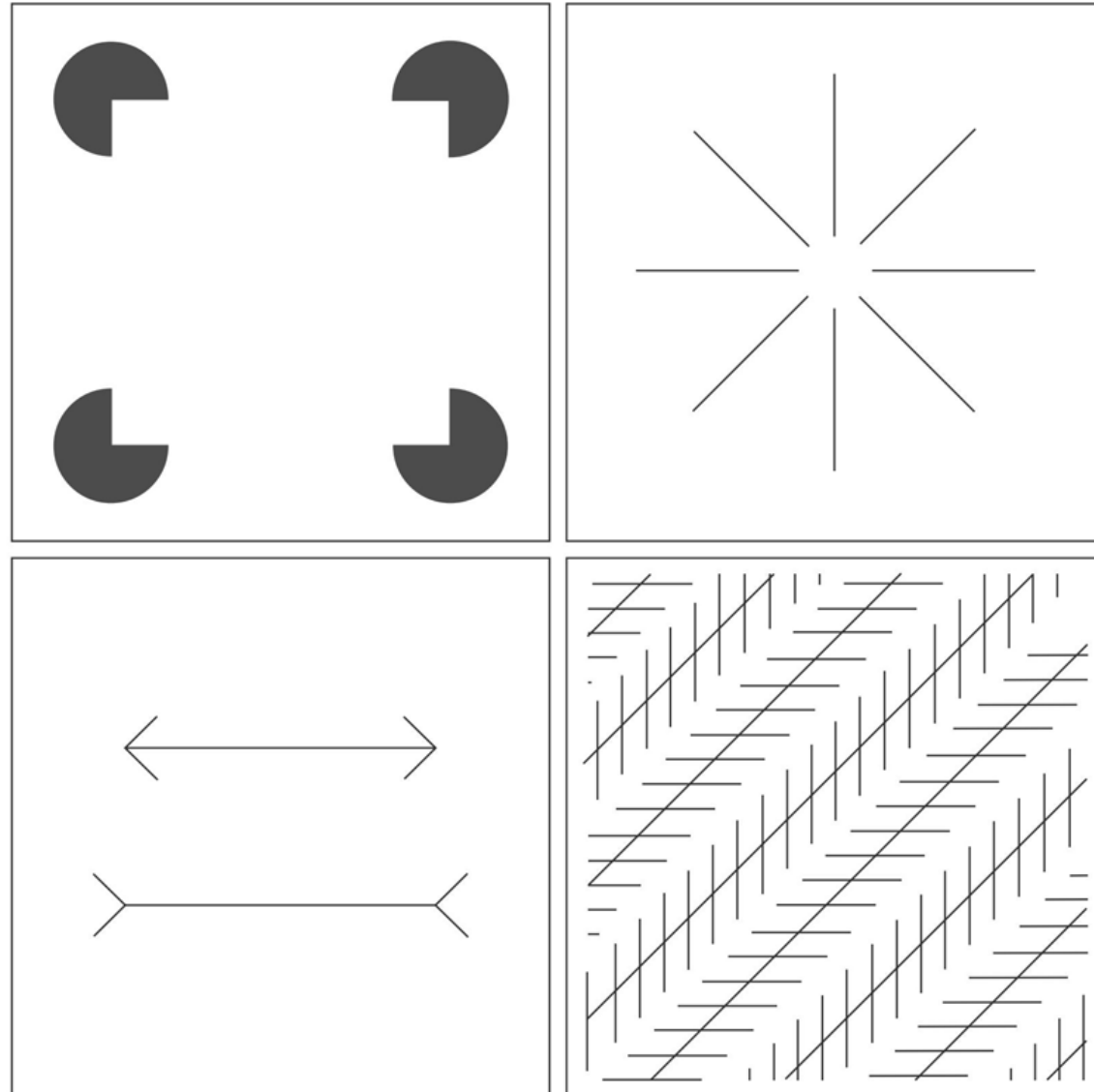


# 人眼的視覺假象(1/2)



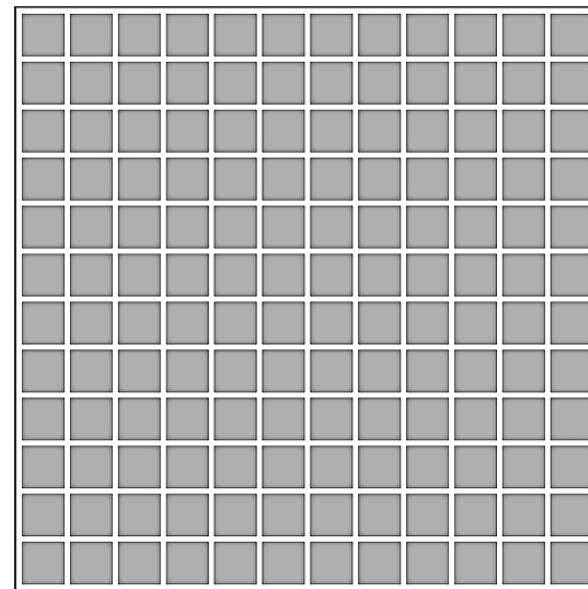
- (A) 1234 都不一樣
- (B) 12 一樣 34 一樣
- (C) 1234 都一樣

## 人眼的視覺假象(2/2)

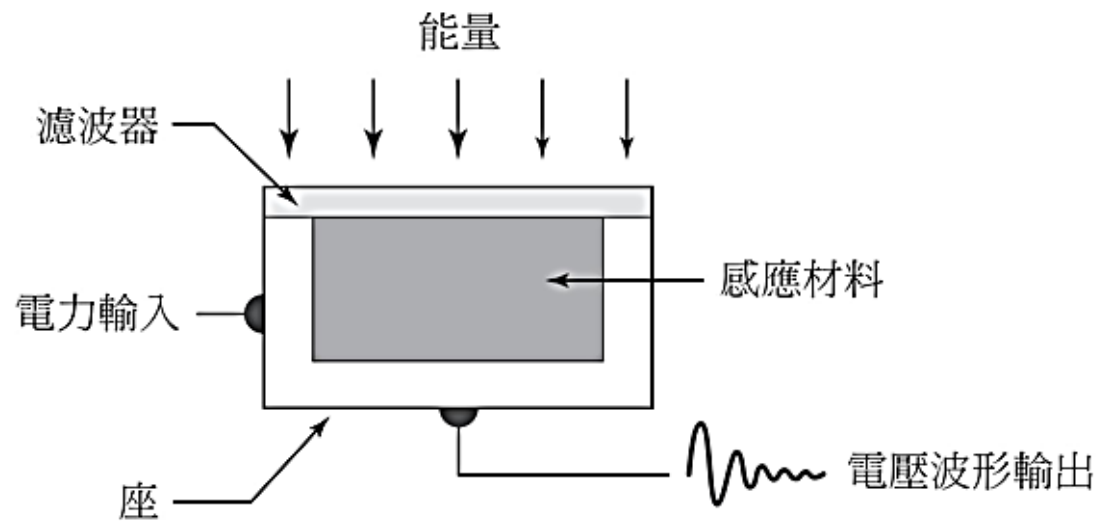


## 2.2 影像的感應與擷取

面影像感測器

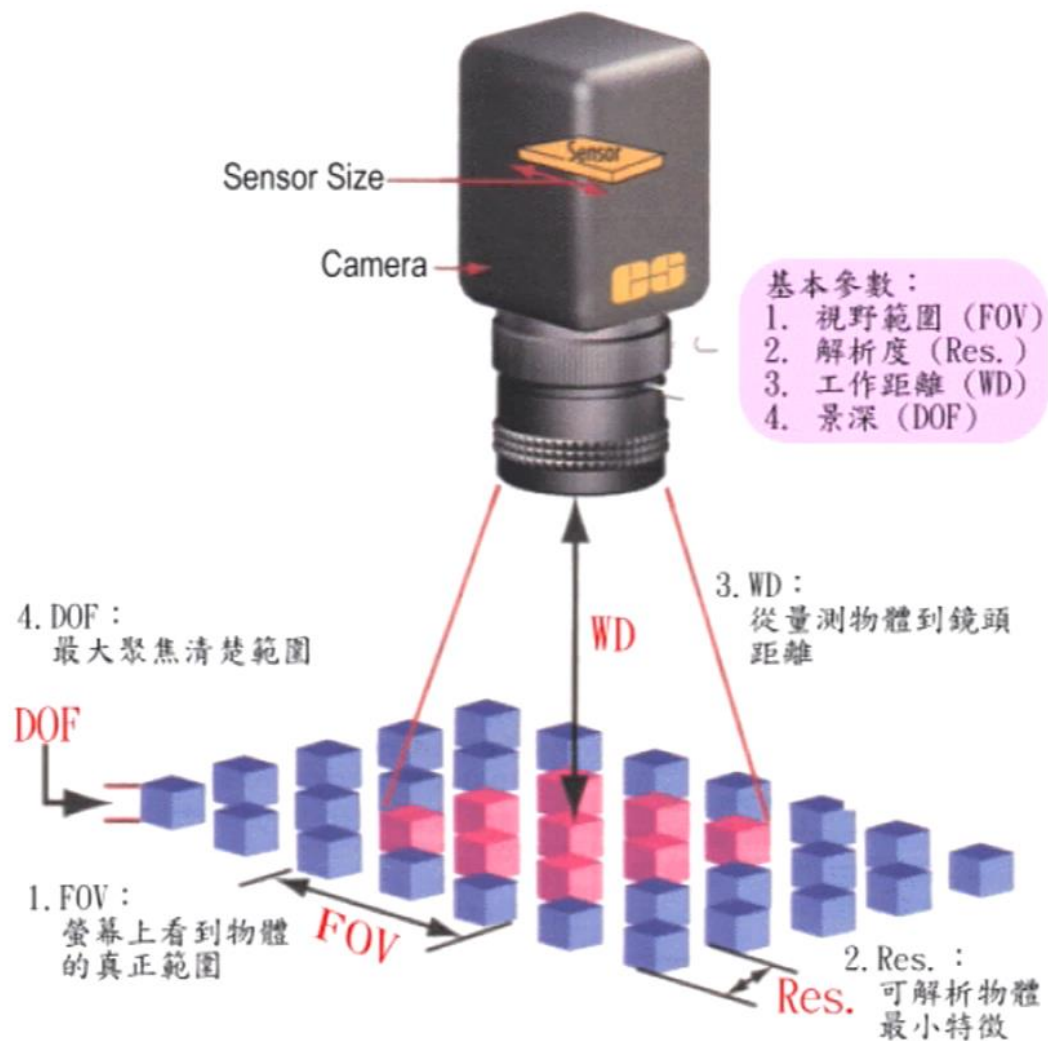


線影像感測器



# 面影像感測器

## ➤ 數位相機的 CMOS、CCD



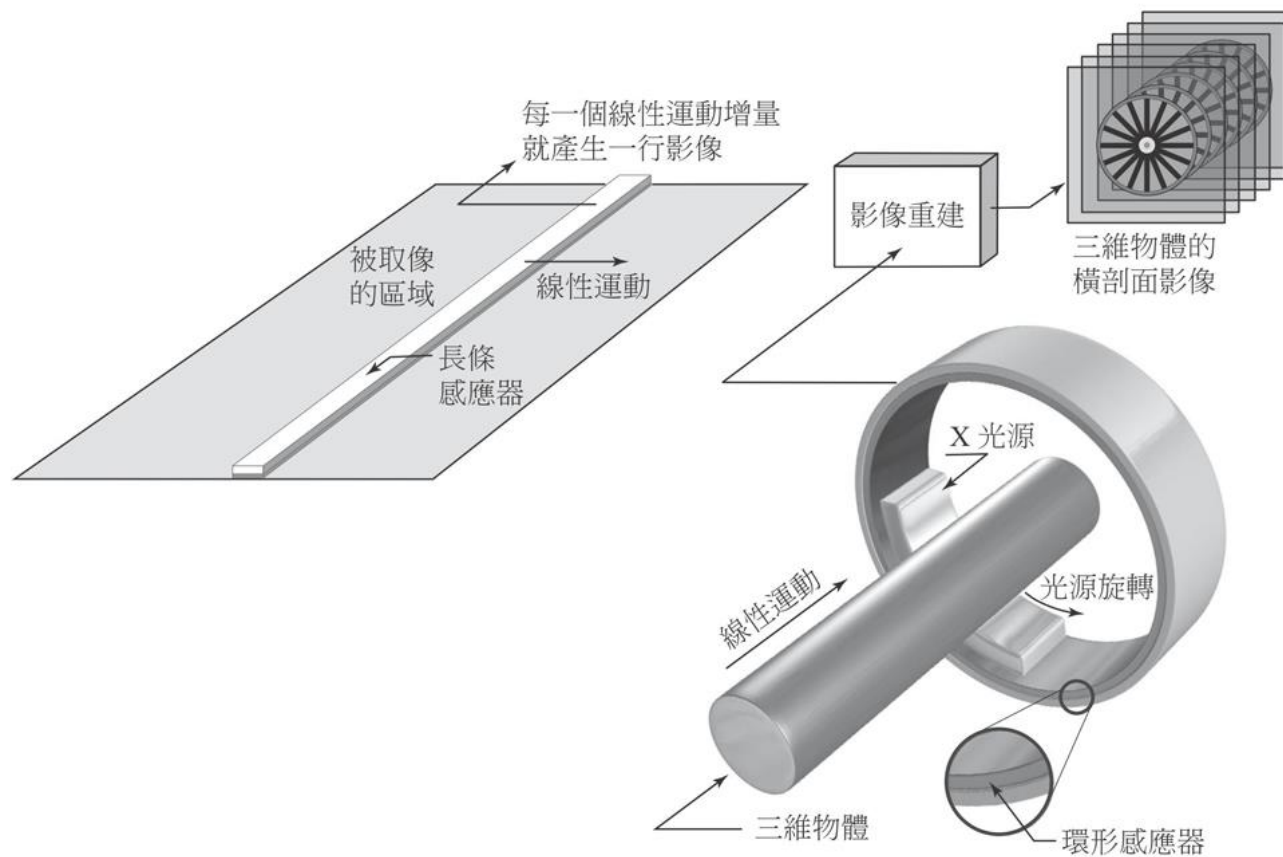


# 線影像感測器

- 刷載具
- 無人機空中成像
- 核磁共振MRI
- PET



@SOGI.com.tw



a b

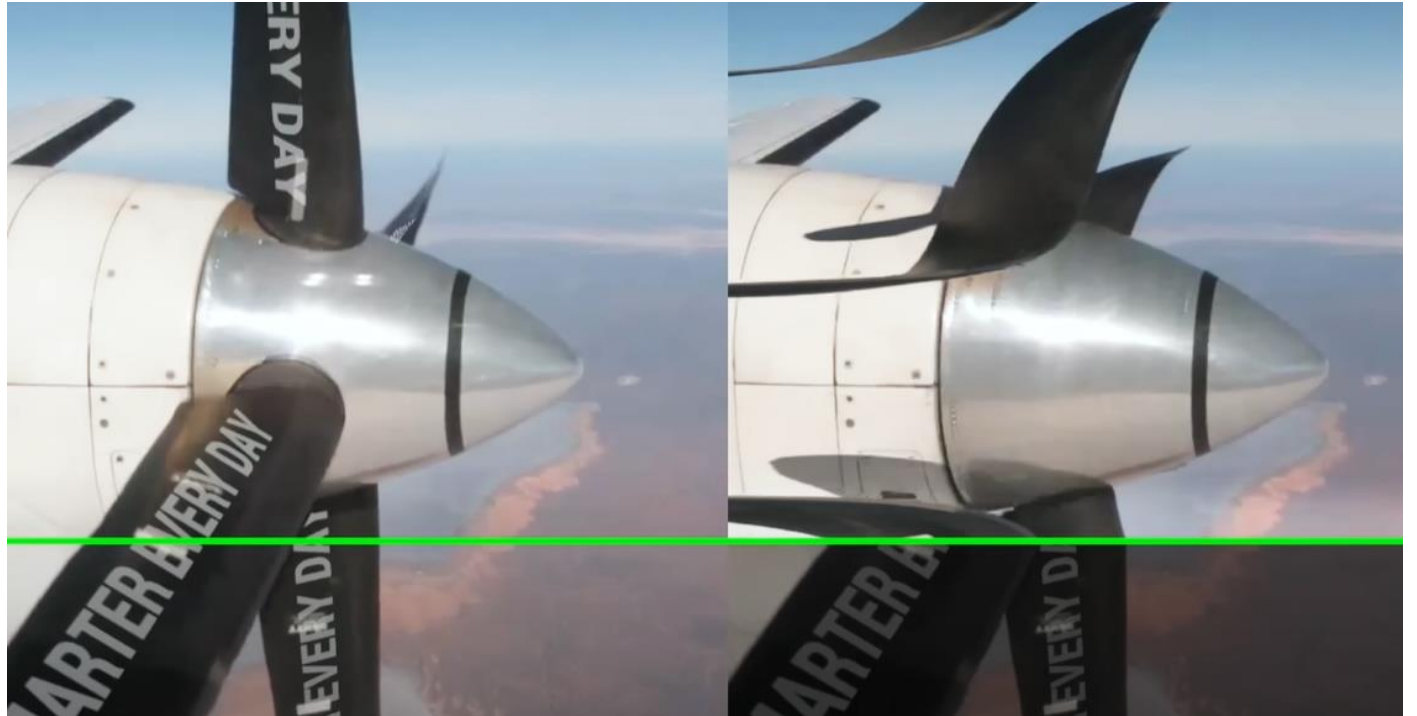
圖 2.12

(a) 用一個長條線型感應器的影像擷取；(b) 用一個圓形長條感應器的影像擷取。

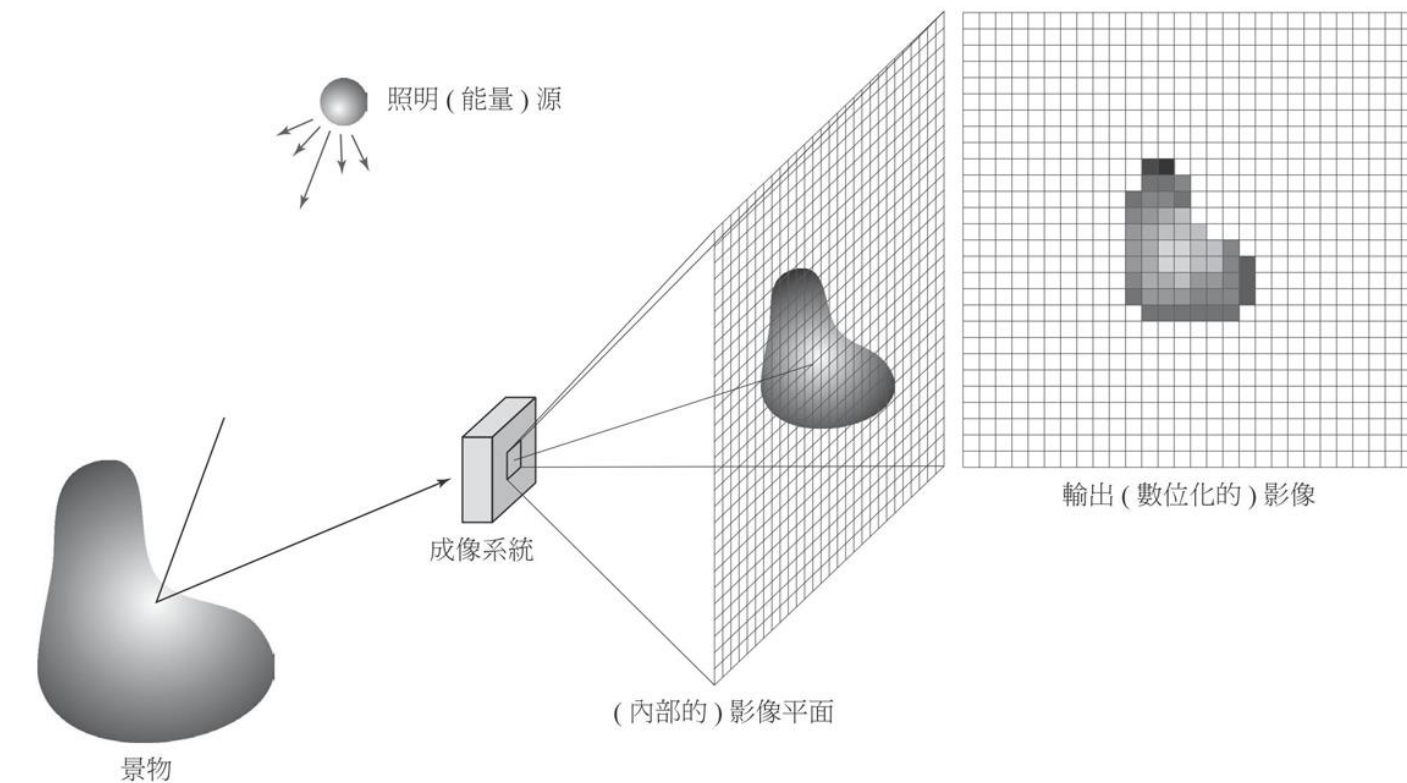
# Rolling Shutter Effect

<https://www.youtube.com/watch?v=dNVtMmLlnoE&t=33s>

- 面影像感測器 (Area sensor) 也會有



# 2D影像形成模型



a  
b c d e

圖 2.13 一個數位影像擷取的例子：(a) 照明 ( 能量 ) 源；(b) 一個景物；(c) 成像系統；(d) 將景物投影到影像平面上；(e) 數位化的影像。

“理論上”

$$0 \leq f(x, y) < \infty$$

亮度值 (intensity) = 照明量 × 反射量

$$f(x, y) = i(x, y)r(x, y)$$

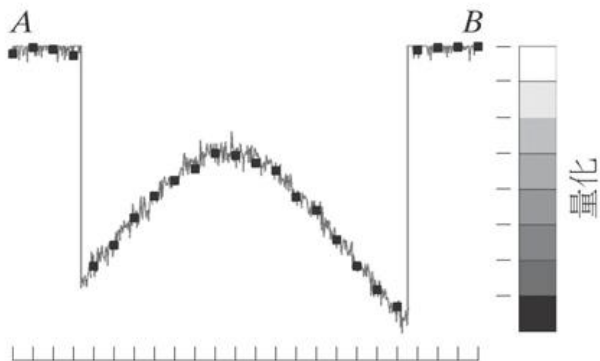
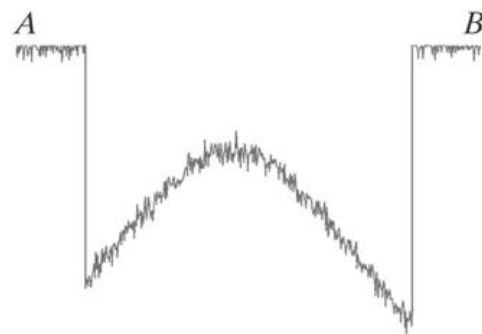
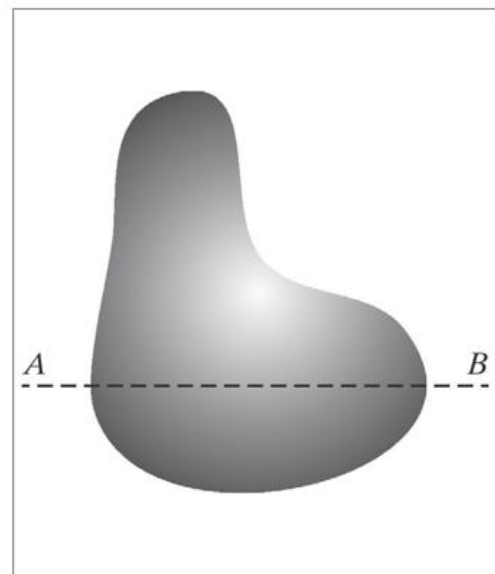
illumination

$$0 \leq i(x, y) < \infty$$

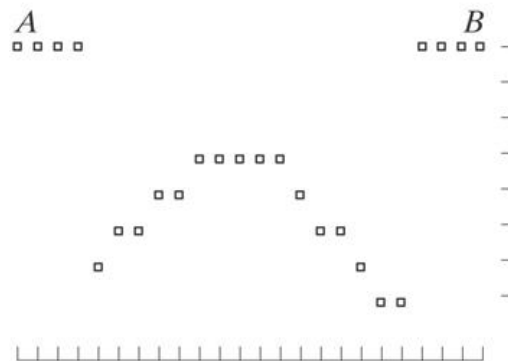
reflectance

$$0 \leq r(x, y) \leq 1$$

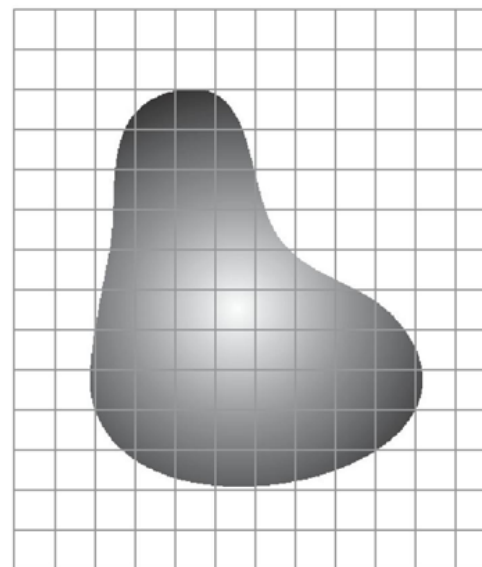
## 2.3 影像的取樣和量化



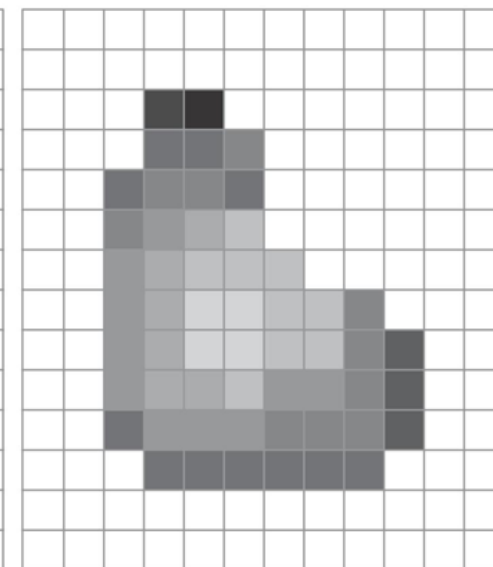
取樣



量化前



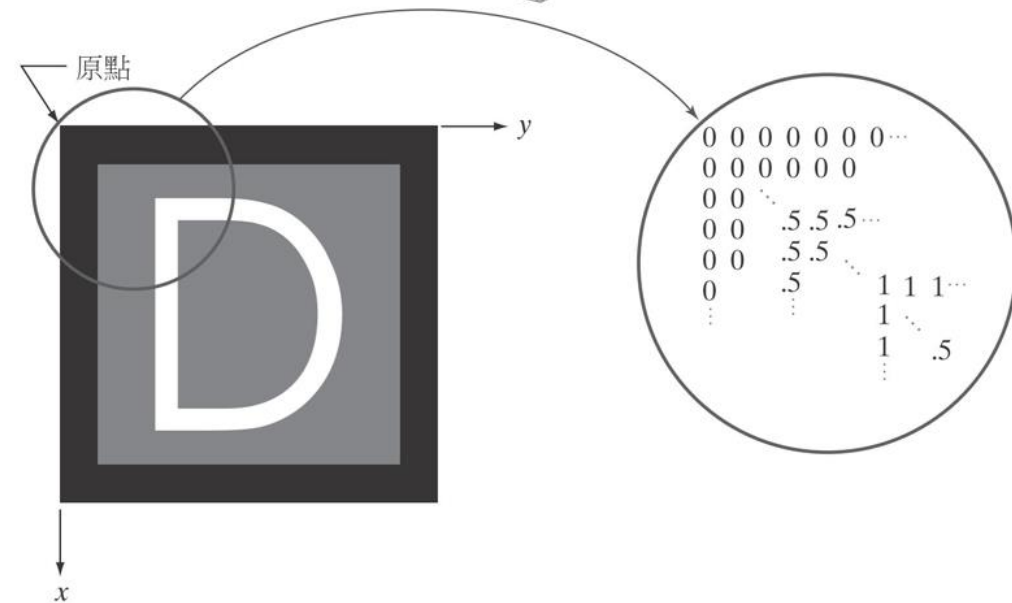
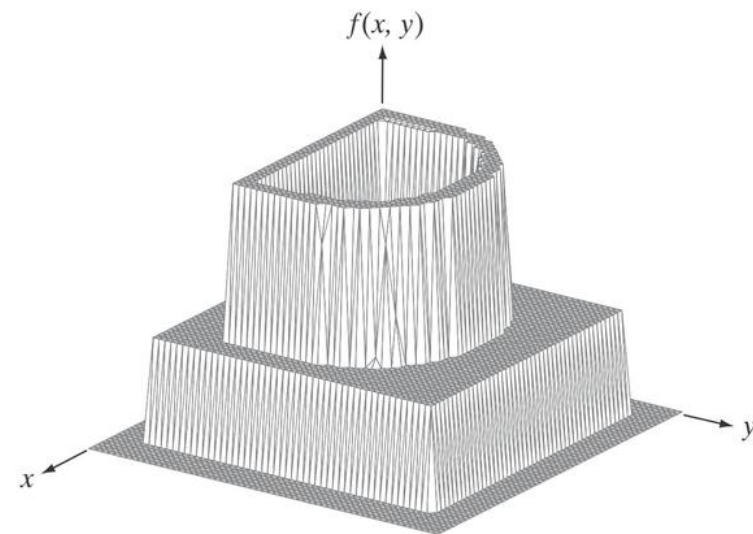
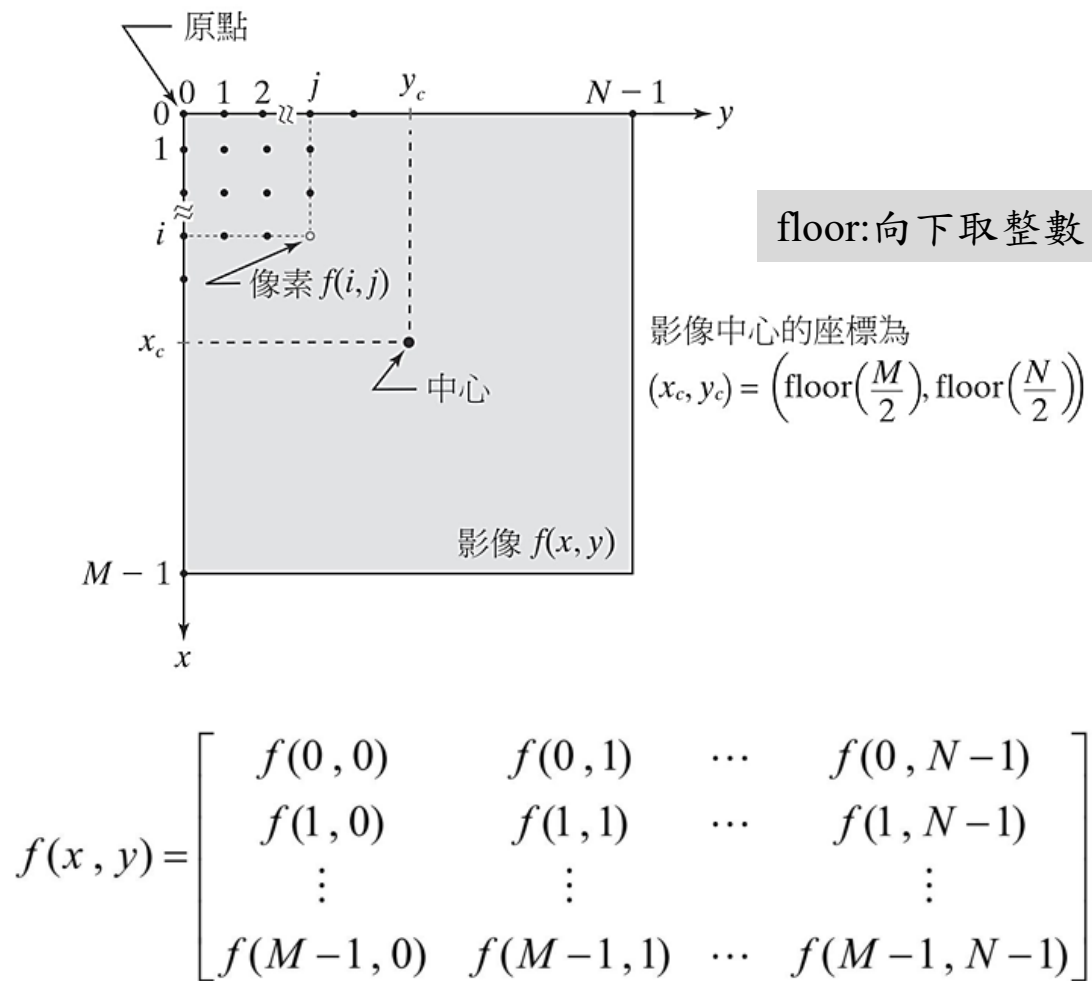
量化後



(8-bit 影像):  $0 \leq f(x, y) \leq 255$



# 數位影像表達

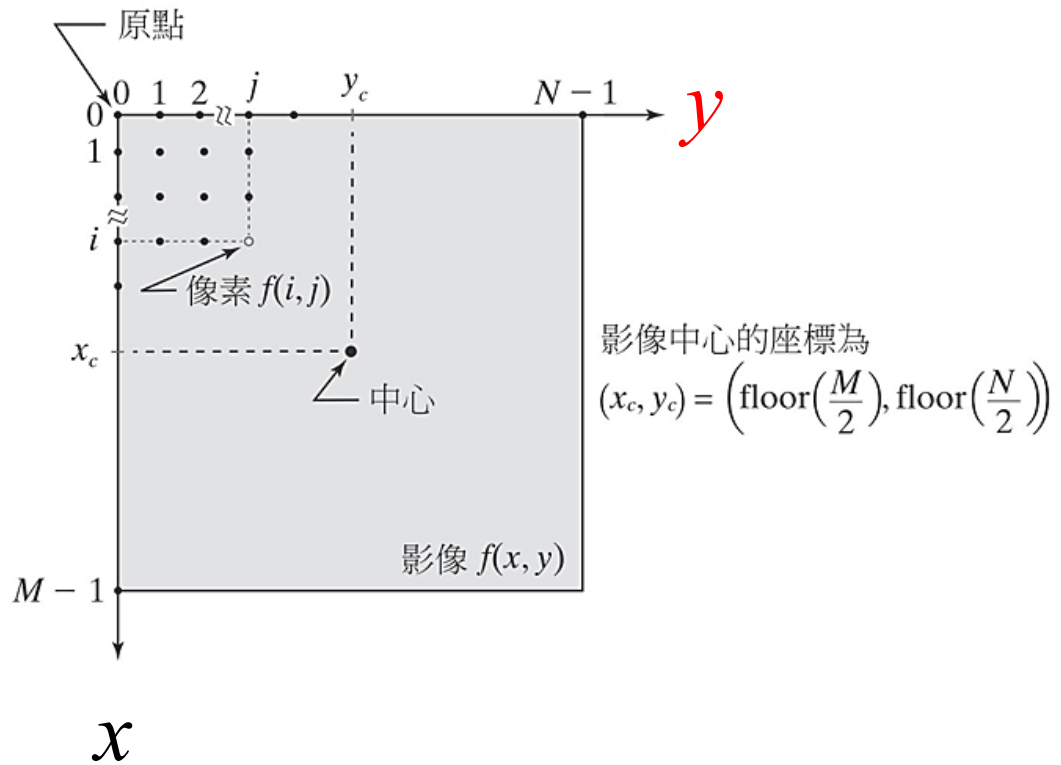


CRT映像管(陰極射線) 原理

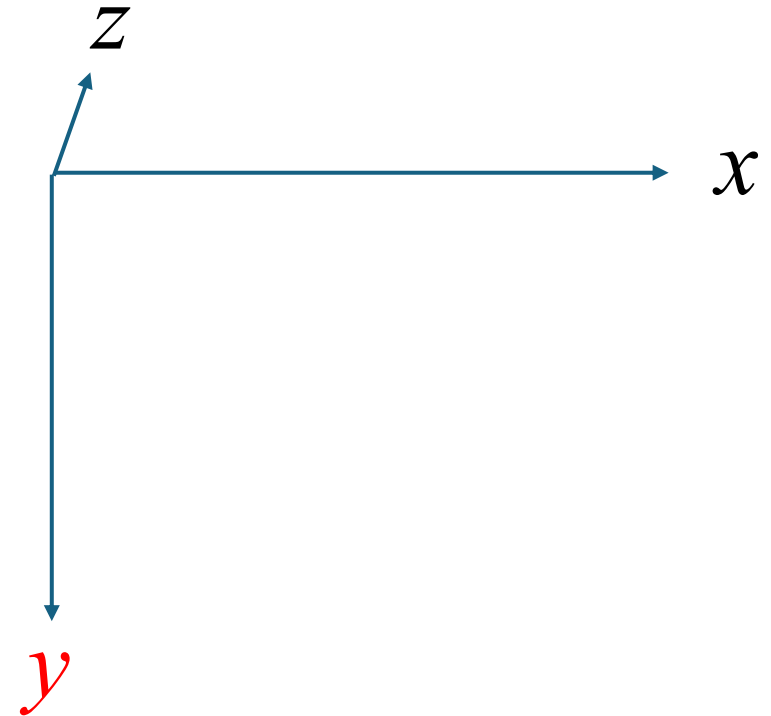
<https://www.youtube.com/watch?v=8ykNPRkmmmbQ>

# 影像座標

影像處理課本



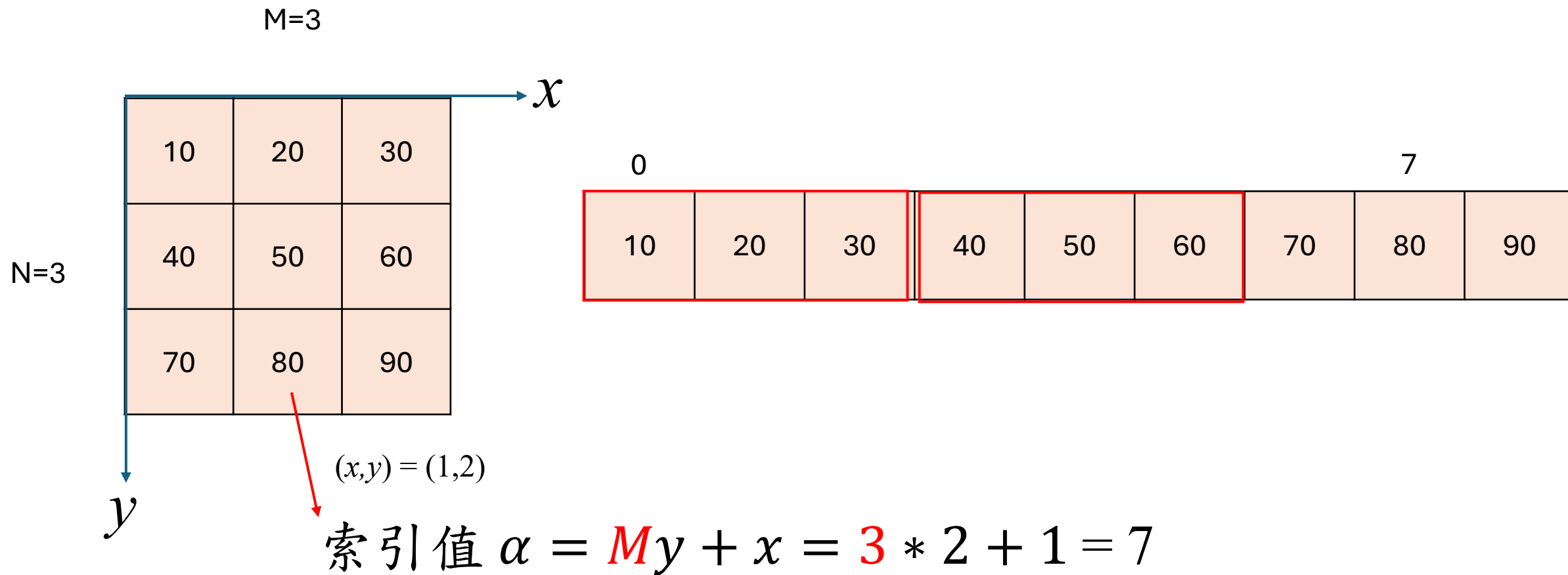
電腦視覺/OpenCV



如果考試的時候有標示不清楚 直接問沒關係



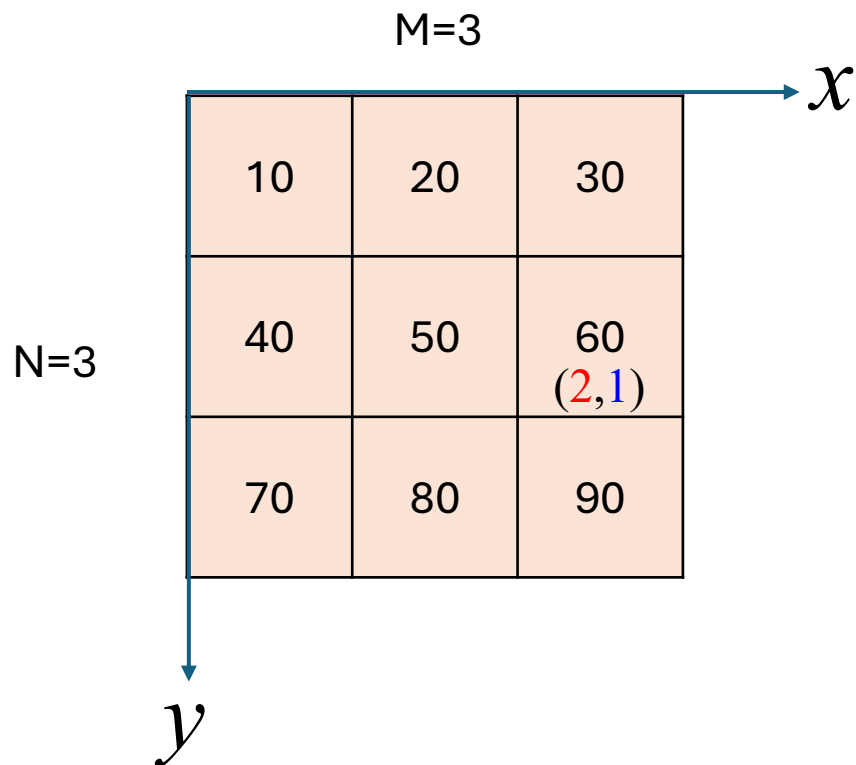
# 影像索引2D







# 影像索引2D



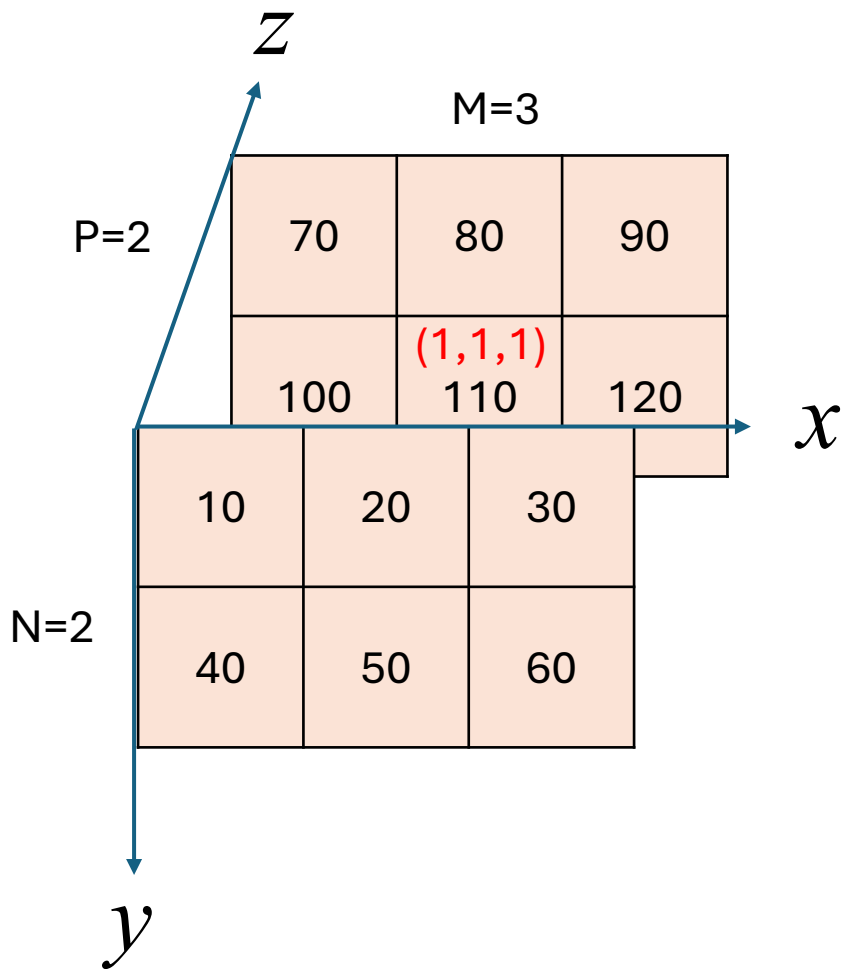
**Q:** ➤ 已知索引值  $\alpha$  求  $(x,y)$   
➤ Given the index value  $\alpha$ , find  $(x,y)$ .

|    |    |    |    |    |              |    |    |    |
|----|----|----|----|----|--------------|----|----|----|
| 0  |    |    |    |    | $\alpha = 5$ |    |    |    |
| 10 | 20 | 30 | 40 | 50 | 60           | 70 | 80 | 90 |

**Sol:** ➤  $x = \alpha \bmod M$                        $x = 5 \bmod 3 = 2$   
➤  $y = \alpha // M$  (取商數)               $y = 5 // 3 = 1$



# 影像索引3D

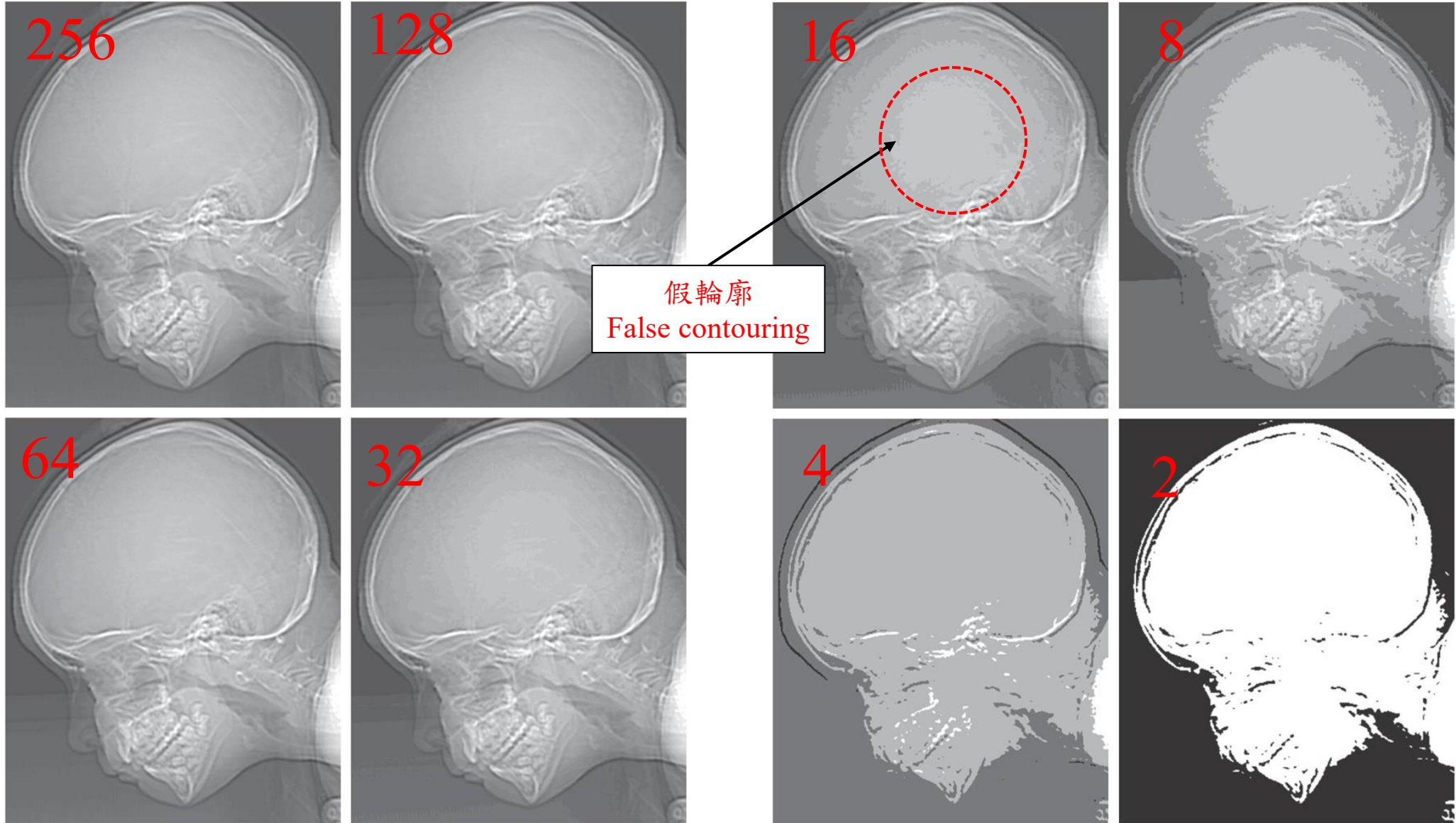


|   |    |    |    |    |    |    |    |    |    |     |     |     |
|---|----|----|----|----|----|----|----|----|----|-----|-----|-----|
| 0 |    |    |    |    |    |    |    |    |    |     |     | 10  |
|   | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 |

索引值  $S = x + My + MNz = x + M(y + Nz)$

- $x = S \bmod MN \bmod M$
- $y = S \bmod MN // M$  (取商數)
- $z = S // MN$  (取商數)

# 強度解析度



# 影像內插(Interpolation)

DPI (Dots Per Inch)

930dpi



300dpi



150dpi



72dpi



鄰近差補



雙線性差補



雙立方差補



a b c

圖 2.25 (a) 縮小到 72 dpi 再用最近鄰插補放大回其原來 930 dpi 大小的影像。此圖與圖 2.21(d) 相同；(b) 縮小再用雙線性插補放大的影像；(c) 與 (b) 同但用雙立方插補。

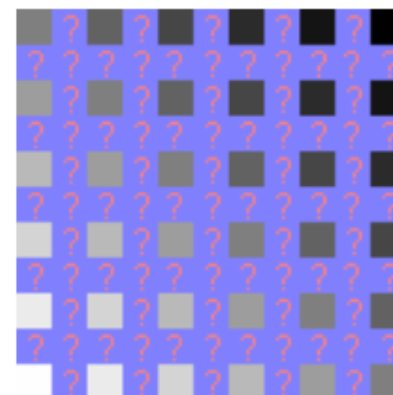
# 影像內插(Interpolation)

- Common interpolation techniques

- Nearest Neighbor (NN)
- Bilinear
- Bicubic



Original



Enlarged



NN

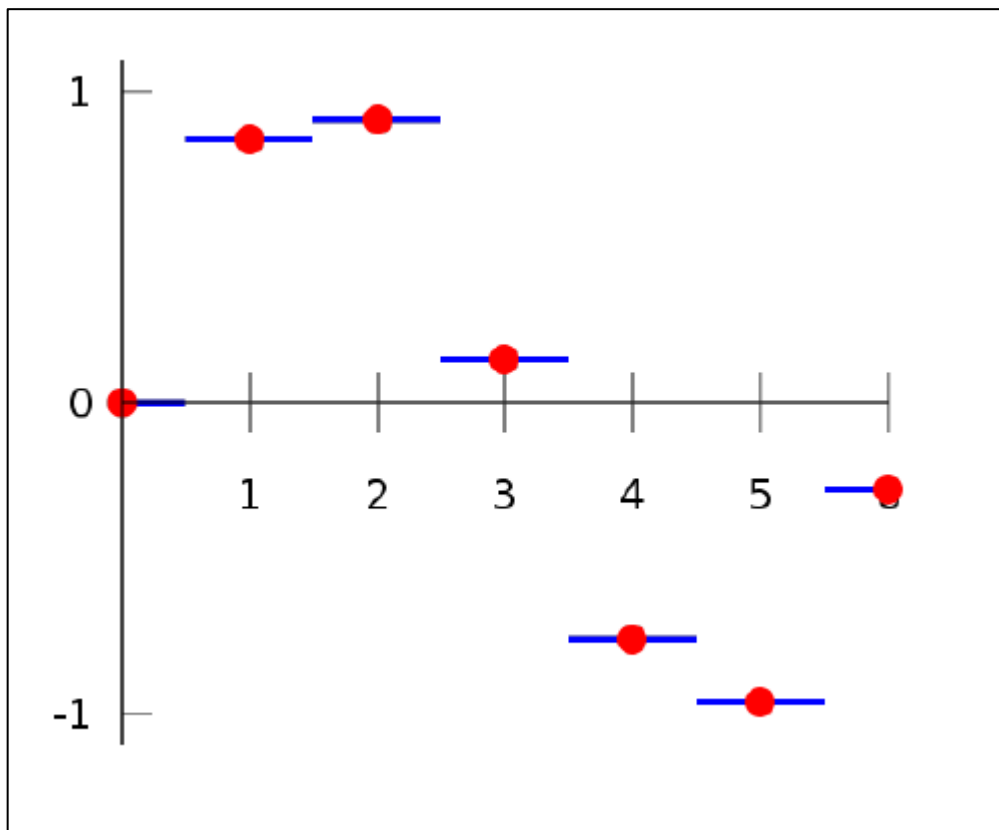


Nearest Neighbor

Bilinear

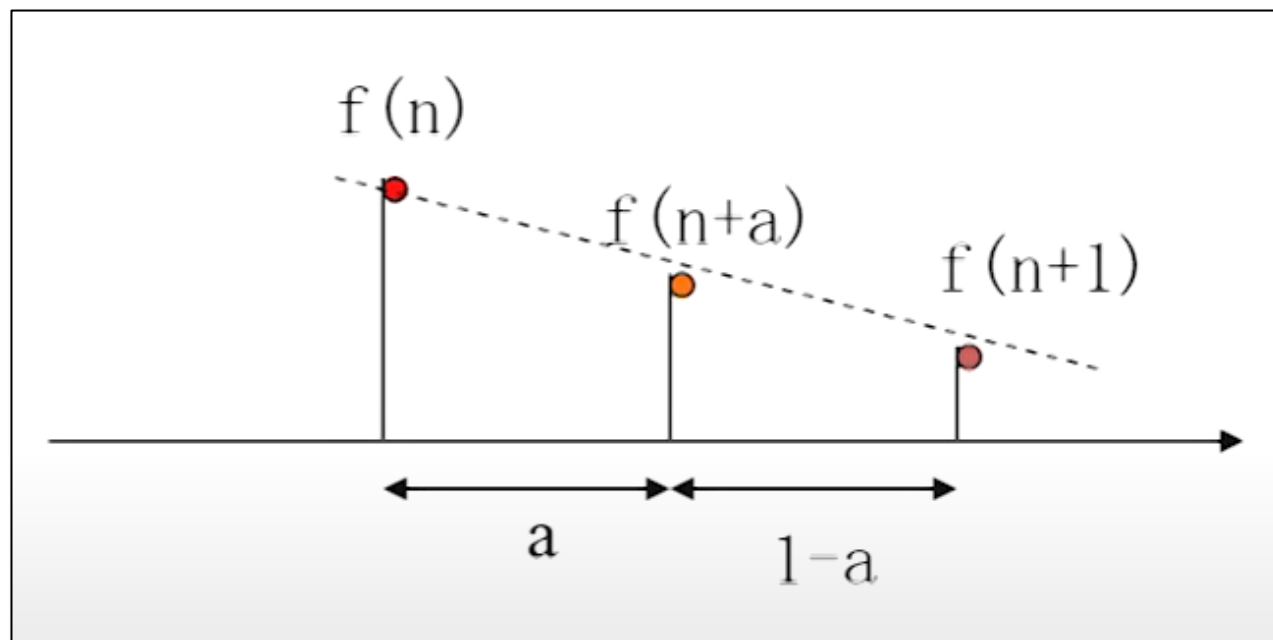
# 1D-影像內插(Interpolation)

1-D最近鄰內插法



1-D線性內插法

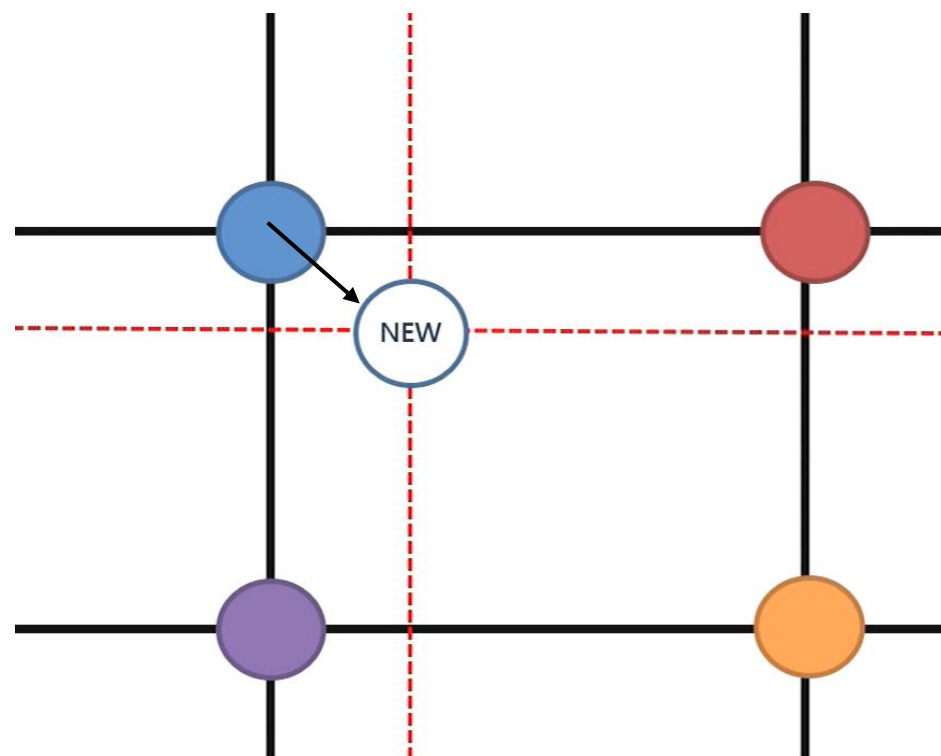
$$f(n+a) = (1-a) \times f(n) + a \times f(n+1)$$





# 2-D最鄰近內插法 (Nearest Neighbor Interpolation)

2-D最鄰近內插法





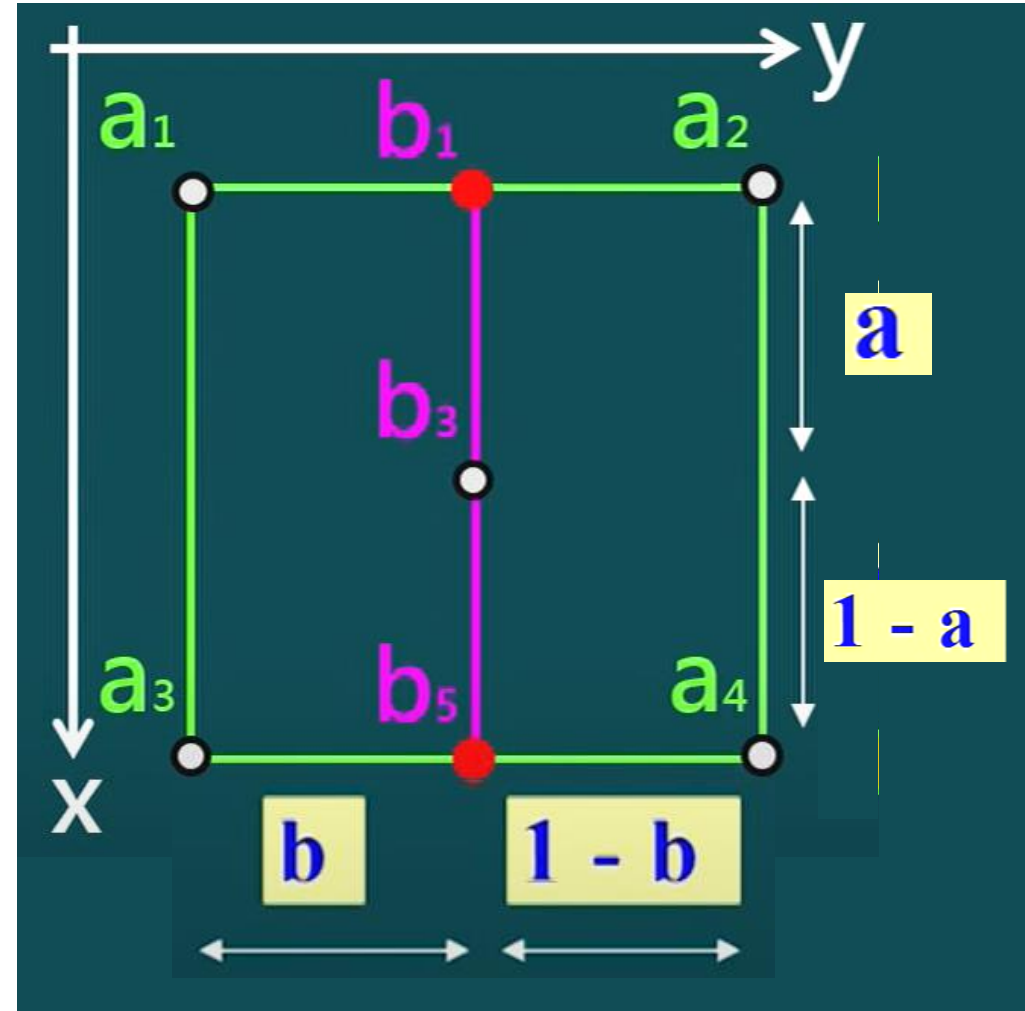
# ★ 2-D 雙線性內插法(Bilinear Interpolation)

1-D線性內插法

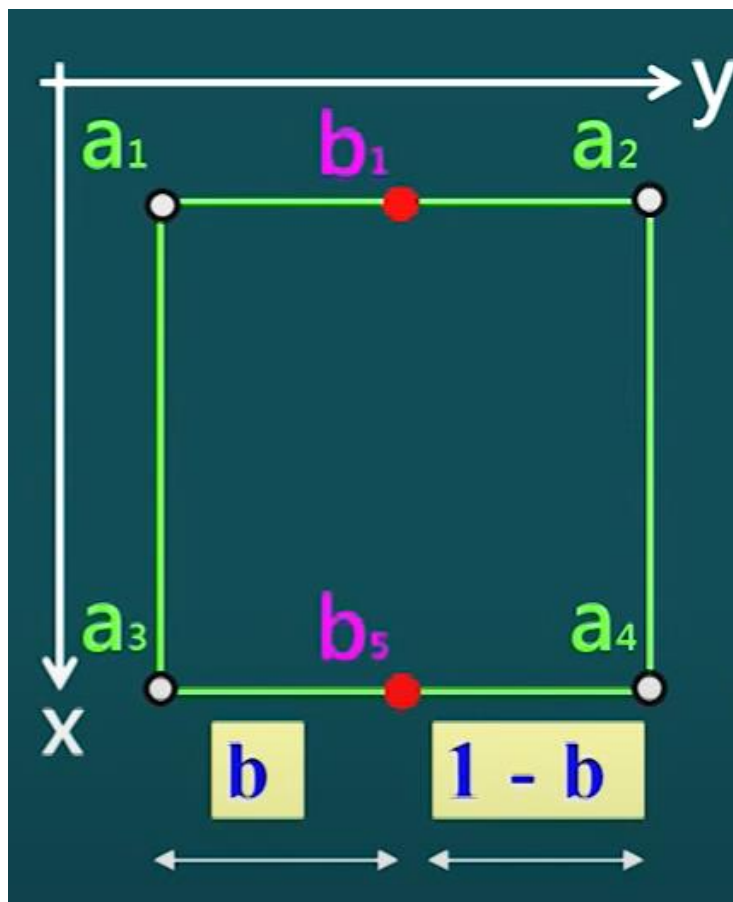
$$f(n + a) = (1 - a) \times f(n) + a \times f(n + 1)$$

2-D線性內插法

$$b_3 = (1 - a) \times [(1 - b) \times a_1 + b \times a_2] \\ + a \times [(1 - b) \times a_3 + b \times a_4]$$



## 2-D 雙線性內插法 推導(1/3):水平方向內插



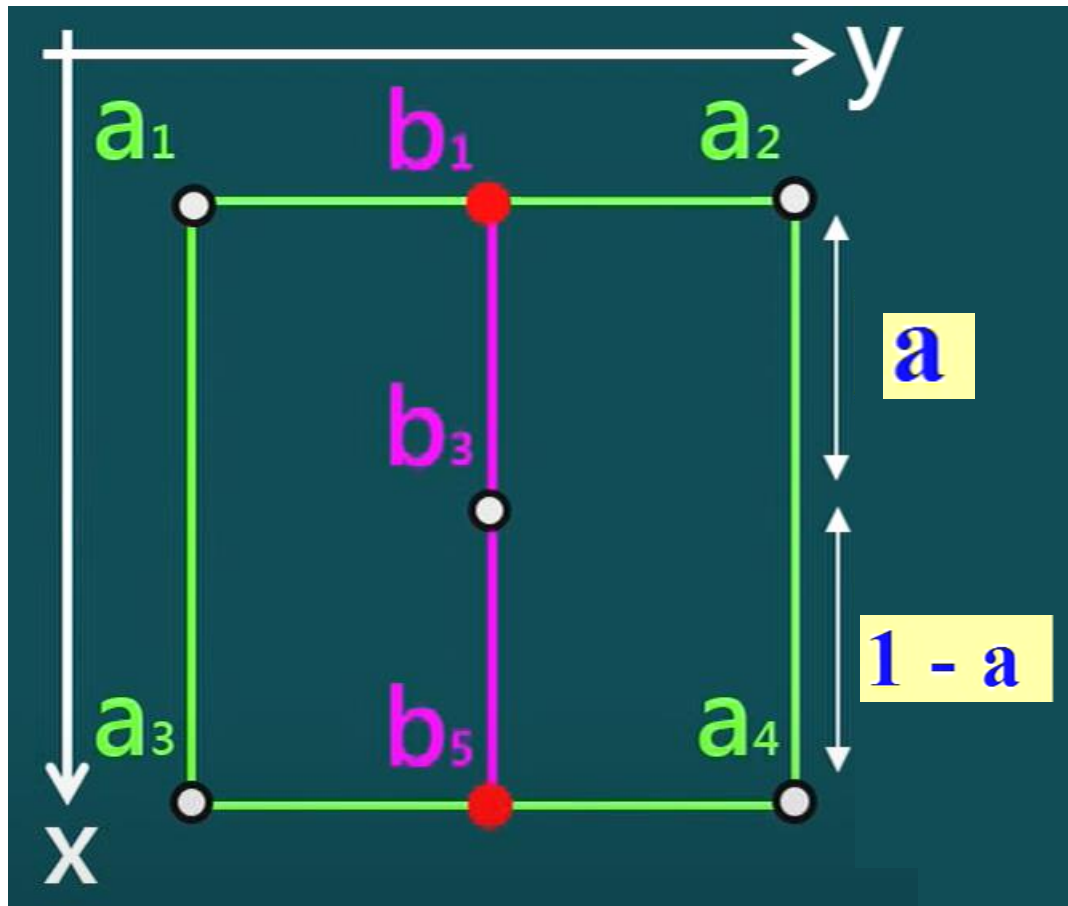
用像素a1、a2 估測出 b1

$$b_1 = (1 - b) \times a_1 + b \times a_2$$

用像素a3、a4 估測出 b5

$$b_5 = (1 - b) \times a_3 + b \times a_4$$

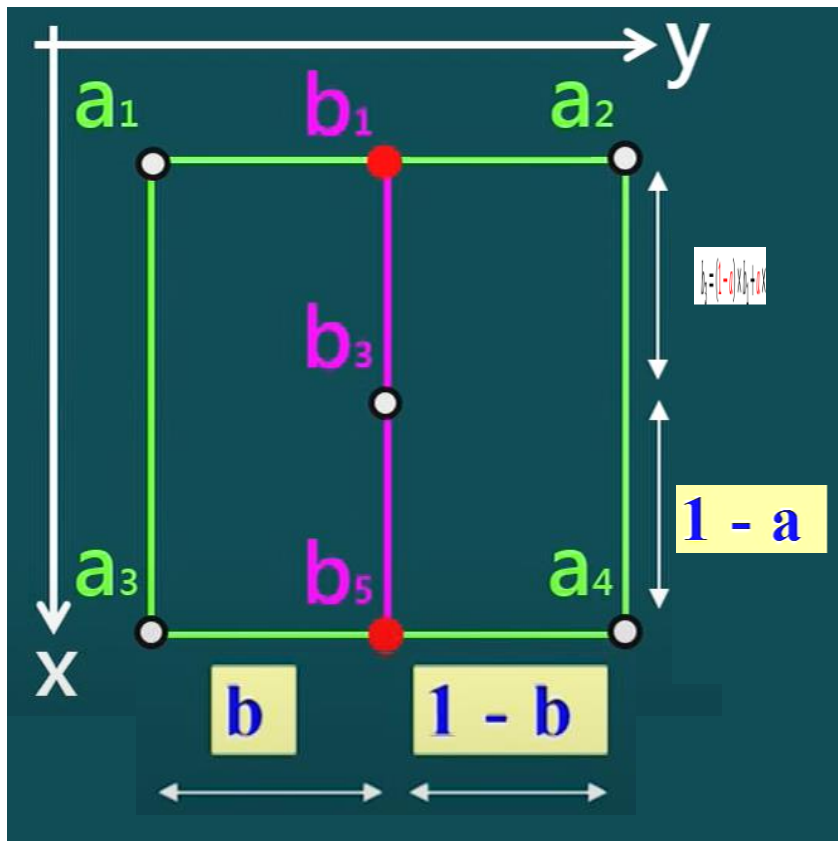
## 2-D 雙線性內插法 推導(2/3):垂直方向內插



用像素  $b_1$ 、 $b_5$  估測出  $b_3$

$$b_3 = (1 - a) \times b_1 + a \times b_5$$

## 2-D 雙線性內插法 推導(3/3):合併



用像素 $a_1$ 、 $a_2$  估測出  $b_1$

$$b_1 = (1 - b) \times a_1 + b \times a_2$$

用像素 $a_3$ 、 $a_4$  估測出  $b_5$

$$b_5 = (1 - b) \times a_3 + b \times a_4$$

用像素 $b_1$ 、 $b_5$  估測出  $b_3$

$$b_3 = (1 - a) \times b_1 + a \times b_5$$

通式

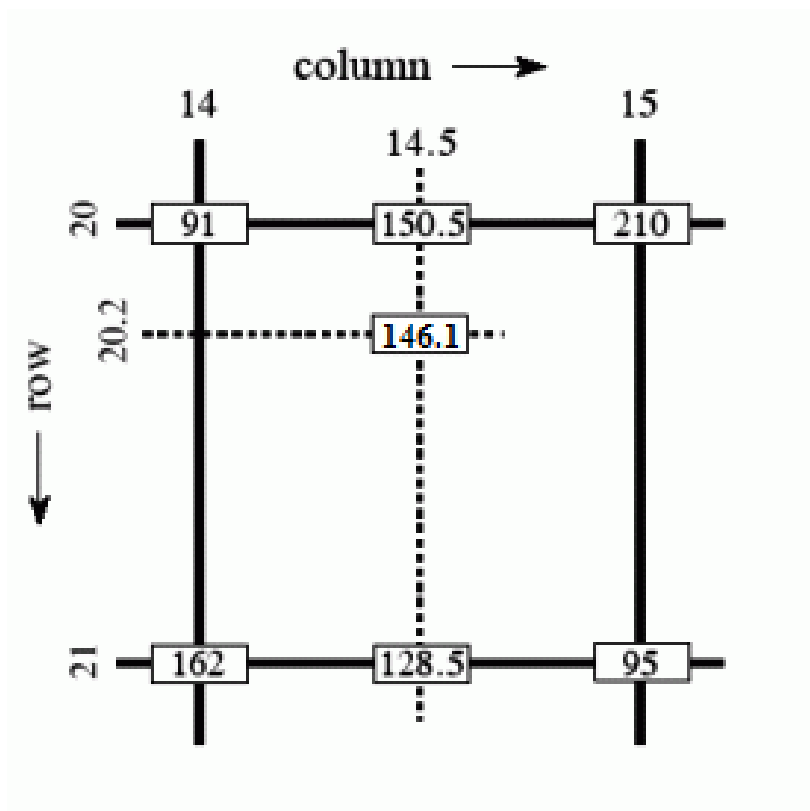
$$b_3 = (1 - a) \times [(1 - b) \times a_1 + b \times a_2] + a \times [(1 - b) \times a_3 + b \times a_4]$$



# 2-D 雙線性內插法 範例Example 1

[https://en.wikipedia.org/wiki/Bilinear\\_interpolation](https://en.wikipedia.org/wiki/Bilinear_interpolation)

**Q:** Using bilinear interpolation, compute the image intensity at point (20.2, 14.5).



$$I_{21,14.5} = \frac{15 - 14.5}{15 - 14} \cdot 91 + \frac{14.5 - 14}{15 - 14} \cdot 210 = 150.5$$

$$I_{21,14.5} = \frac{15 - 14.5}{15 - 14} \cdot 162 + \frac{14.5 - 14}{15 - 14} \cdot 95 = 128.5$$

$$I_{20.2,14.5} = \frac{21 - 20.2}{21 - 20} \cdot 150.5 + \frac{20.2 - 20}{21 - 20} \cdot 128.5 = 146.1$$



## 2-D 雙線性內插法 範例Example 2

➤ 40會在哪?

➤ 內插時A點與10跟40的距離?

$\frac{2}{3}$ 、 $\frac{1}{3}$

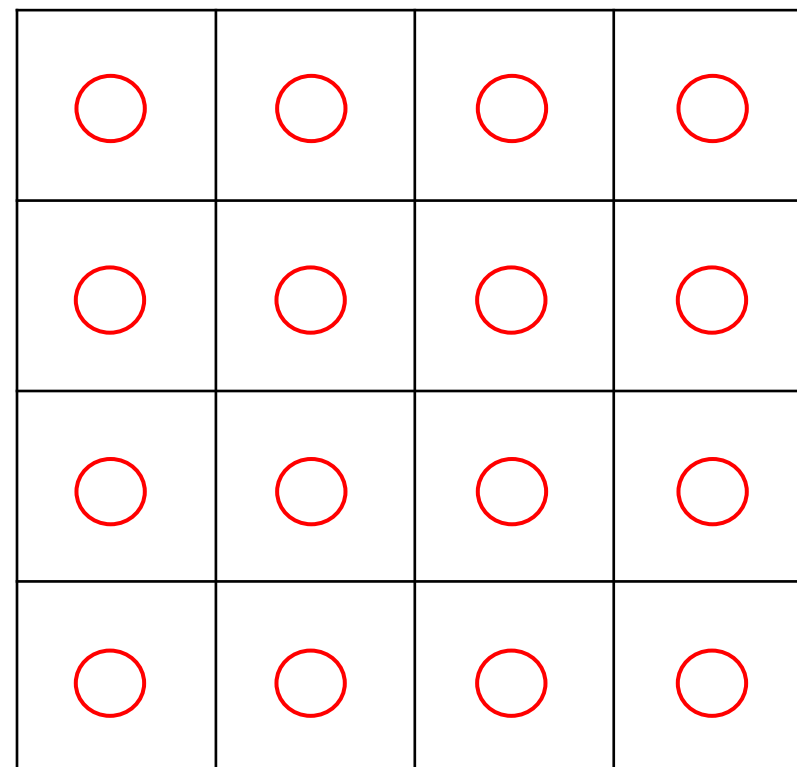
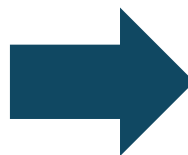
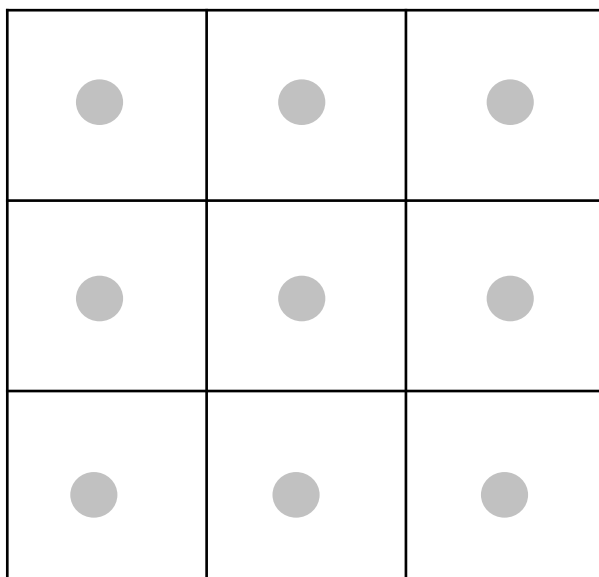
|    |    |    |
|----|----|----|
| 10 | 20 | 30 |
| 40 | 50 | 60 |
| 70 | 80 | 90 |



|    |  |  |    |
|----|--|--|----|
| 10 |  |  | 30 |
| A  |  |  |    |
|    |  |  |    |
| 70 |  |  | 90 |

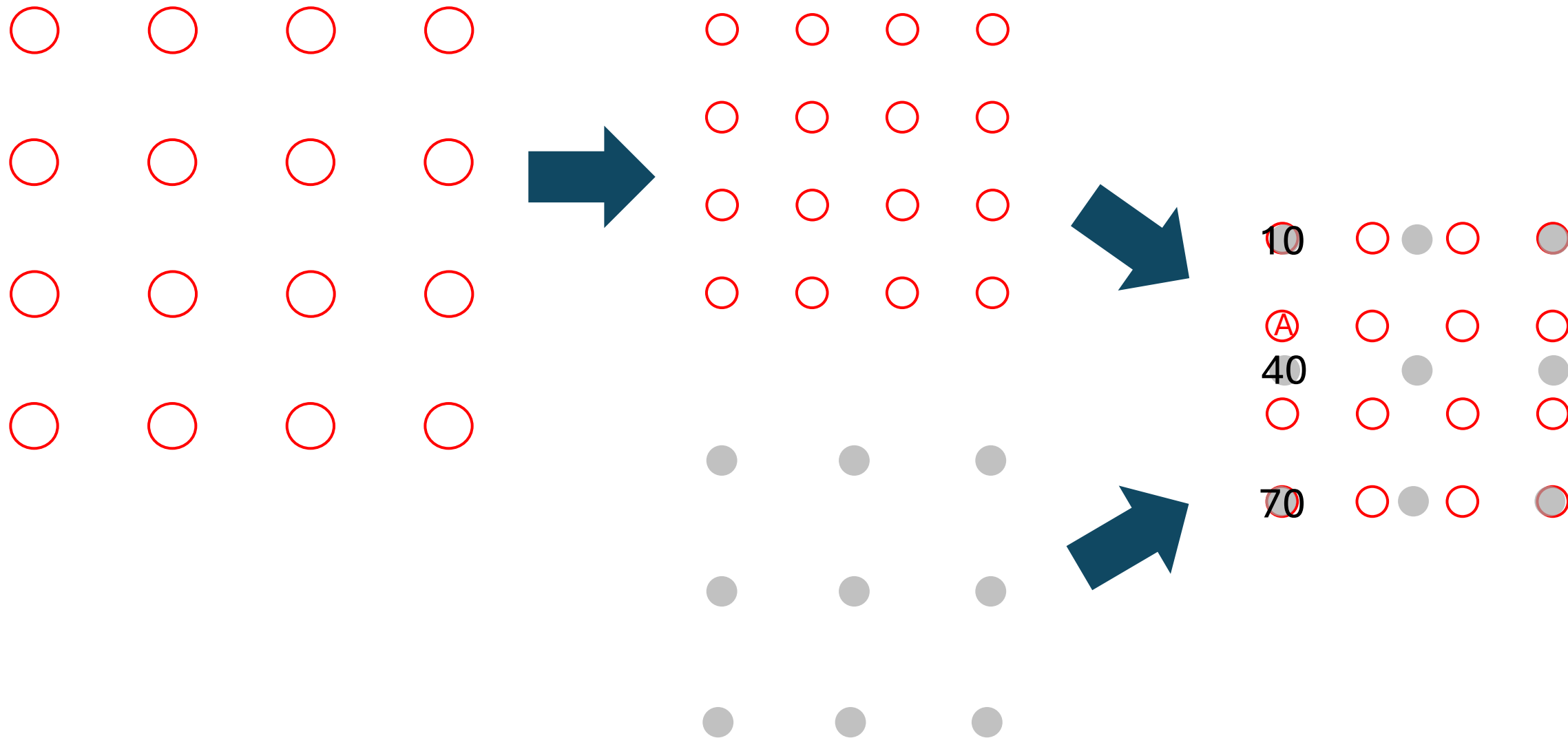
## 2-D 雙線性內插法 範例Example 2

抓像素中心點

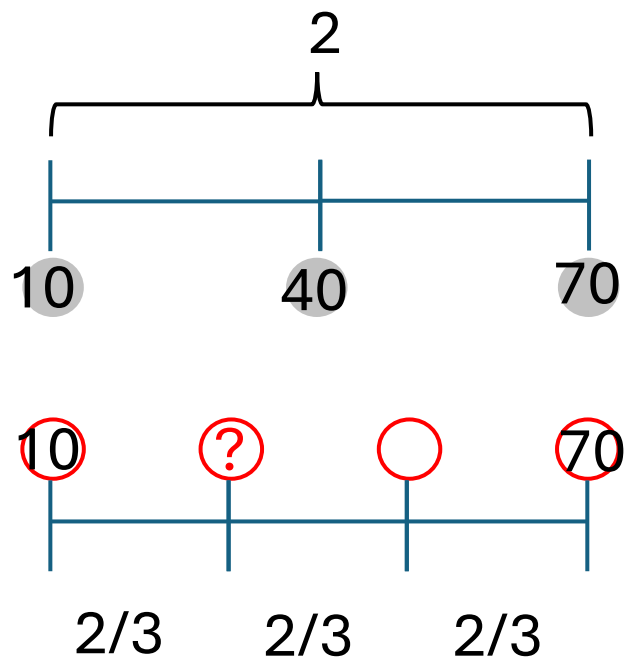




# 四個角落中心點對齊



# 計算縮放比例



3個像素總長=2

4個像素總長=3

總長度2要平分3個間距

$$1\text{個間距長度}=\frac{2}{3}$$



## Step1:計算縮放比例

- 原始圖像尺寸:寬度 $w_1 = 3$  , 高度 $h_1 = 3$
- 目標圖像尺寸:寬度 $w_2 = 4$  , 高度 $h_2 = 4$

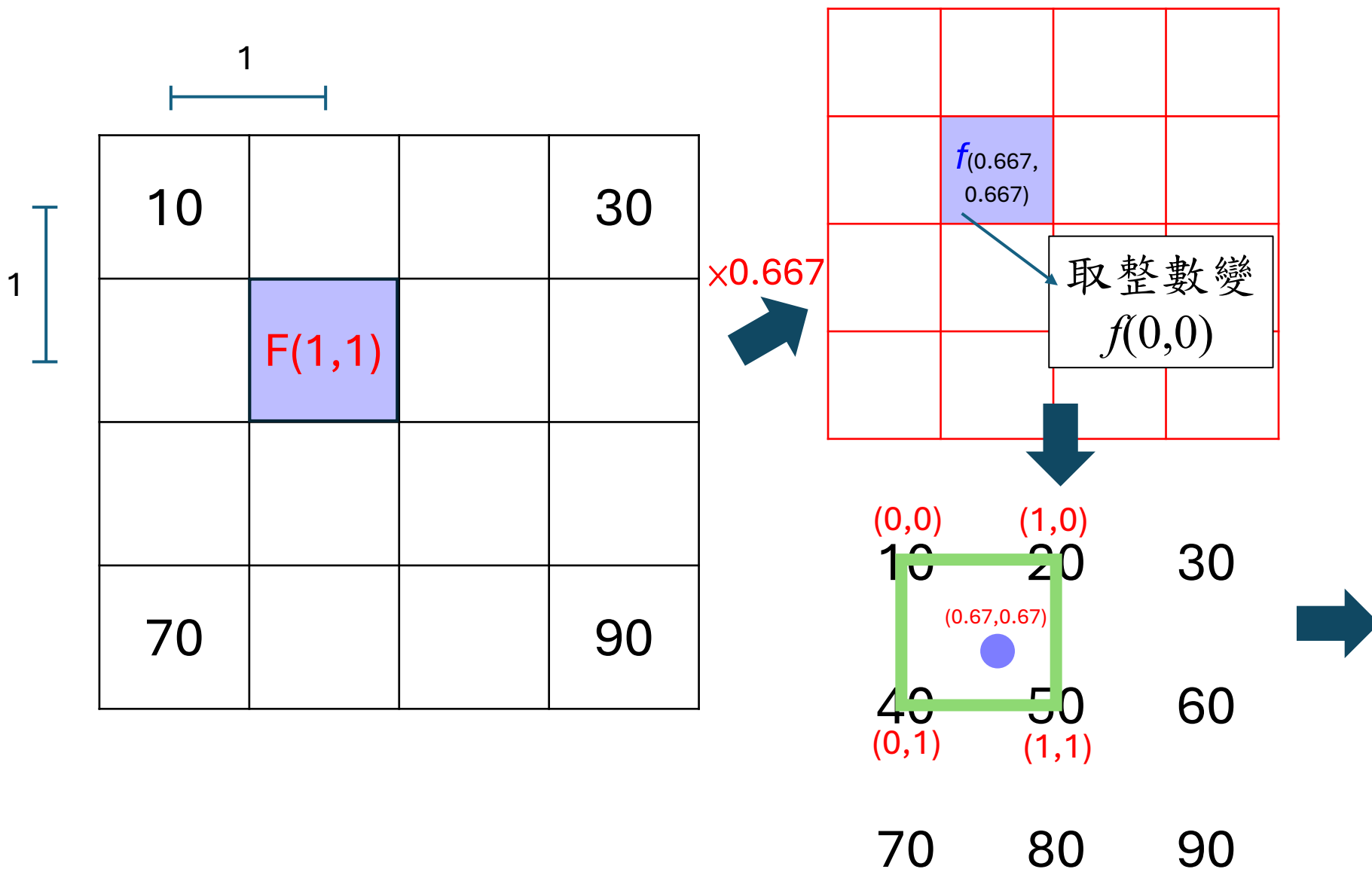
$$S_x = \frac{w_1 - 1}{w_2 - 1} = \frac{3 - 1}{4 - 1} = \frac{2}{3} \approx 0.667$$

$$S_y = \frac{h_1 - 1}{h_2 - 1} = \frac{2}{3} \approx 0.667$$

這個就是內差時的距離

F: 目標圖  
f: 原圖

## Step2:找鄰近四個點

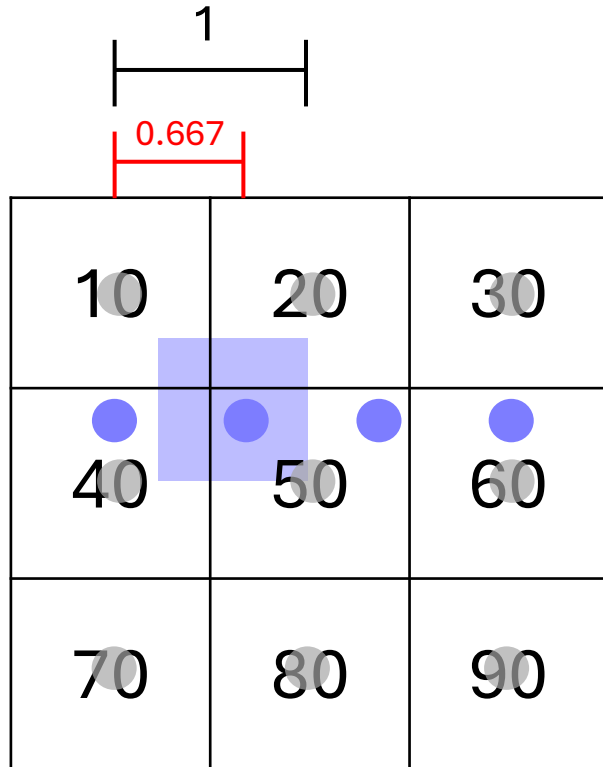


- $a_1 = f(x_1, y_1)$
- $a_2 = f(x_1 + 1, y_1)$
- $a_3 = f(x_1, y_1 + 1)$
- $a_4 = f(x_1 + 1, y_1 + 1)$

座標各減1 = 鄰近4個點

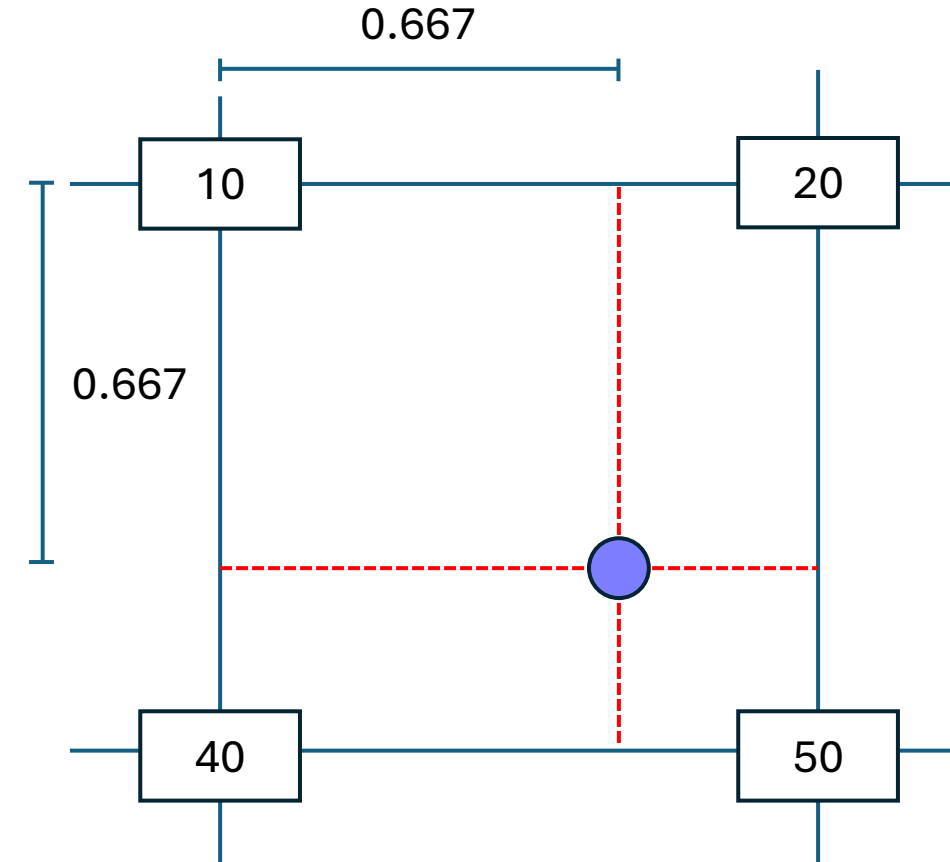
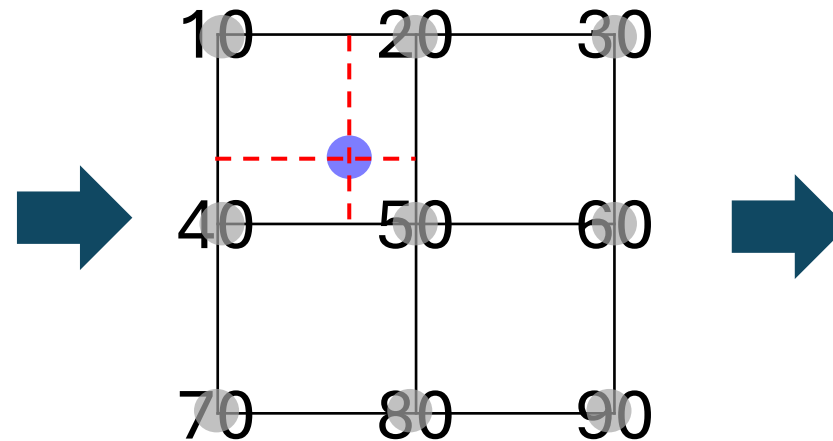
|          |          |    |
|----------|----------|----|
| $f(0,0)$ | $f(1,0)$ | 30 |
| 10       | 20       |    |
| $f(0,1)$ | $f(1,1)$ | 60 |
| 40       | 50       |    |
| 70       | 80       | 90 |

# Step3:計算內插值(1)



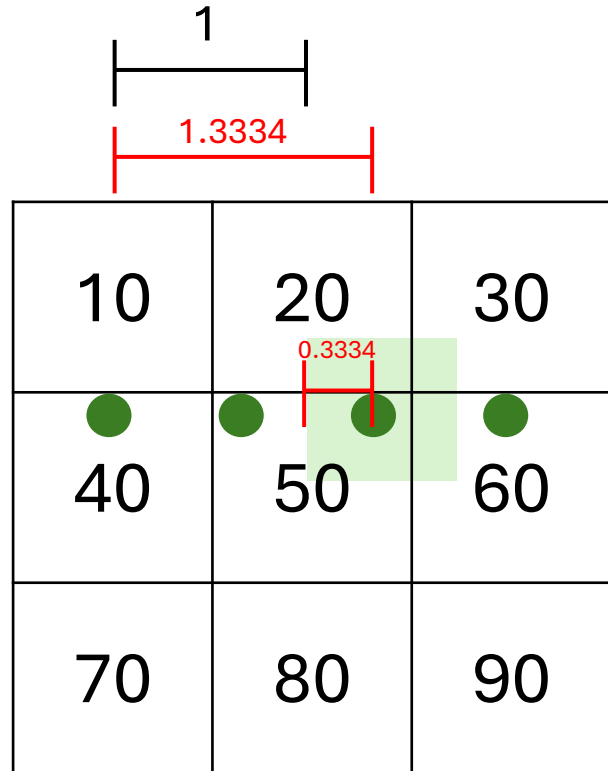
$$0.6667 * 1 = 0.6667$$

$$0.6667 * 1 = 0.6667$$



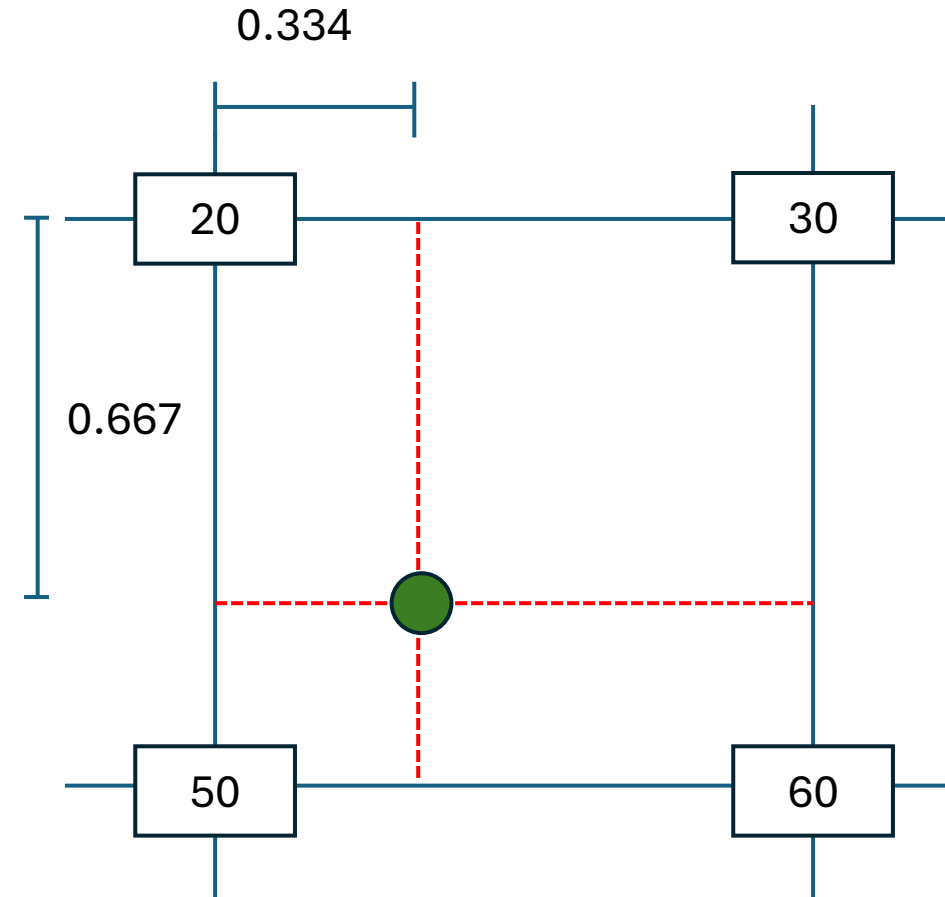
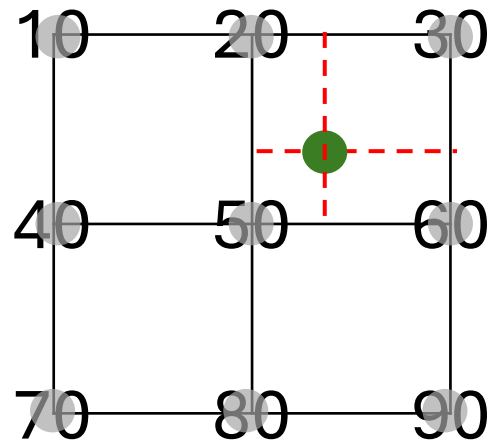
$$F(1,1) = (1-0.6667) \times [(1-0.6667) \times 10 + 0.6667 \times 40] + 0.6667 \times [(1-0.6667) \times 20 + 0.6667 \times 50] = 36.6672$$

## Step3:計算內插值(2)



$$0.6667 * 2 = 1.3334 \quad \text{扣除整數} = 0.3334$$

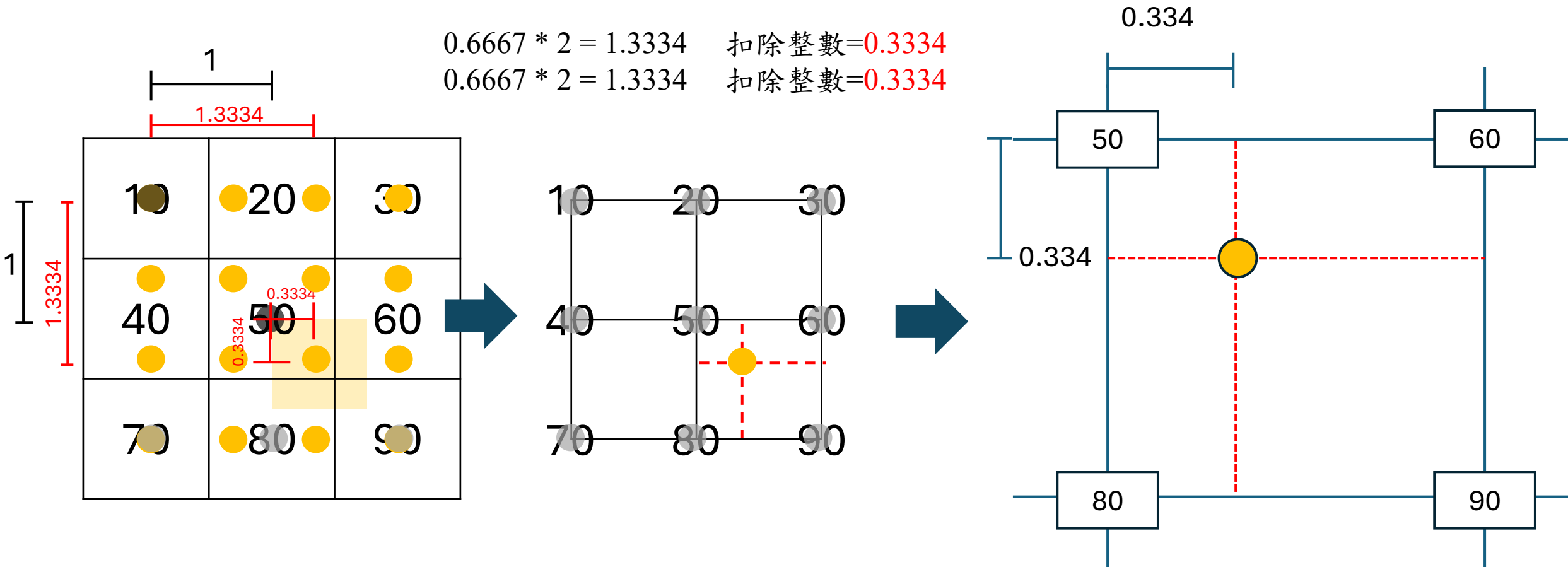
$$0.6667 * 1 = 0.667$$



$$F(2,1) = (1-0.3334) \times [(1-0.6667) \times 20 + 0.6667 \times 50] +$$

$$0.3334 \times [(1-0.6667) \times 30 + 0.6667 \times 60] = 43.3352$$

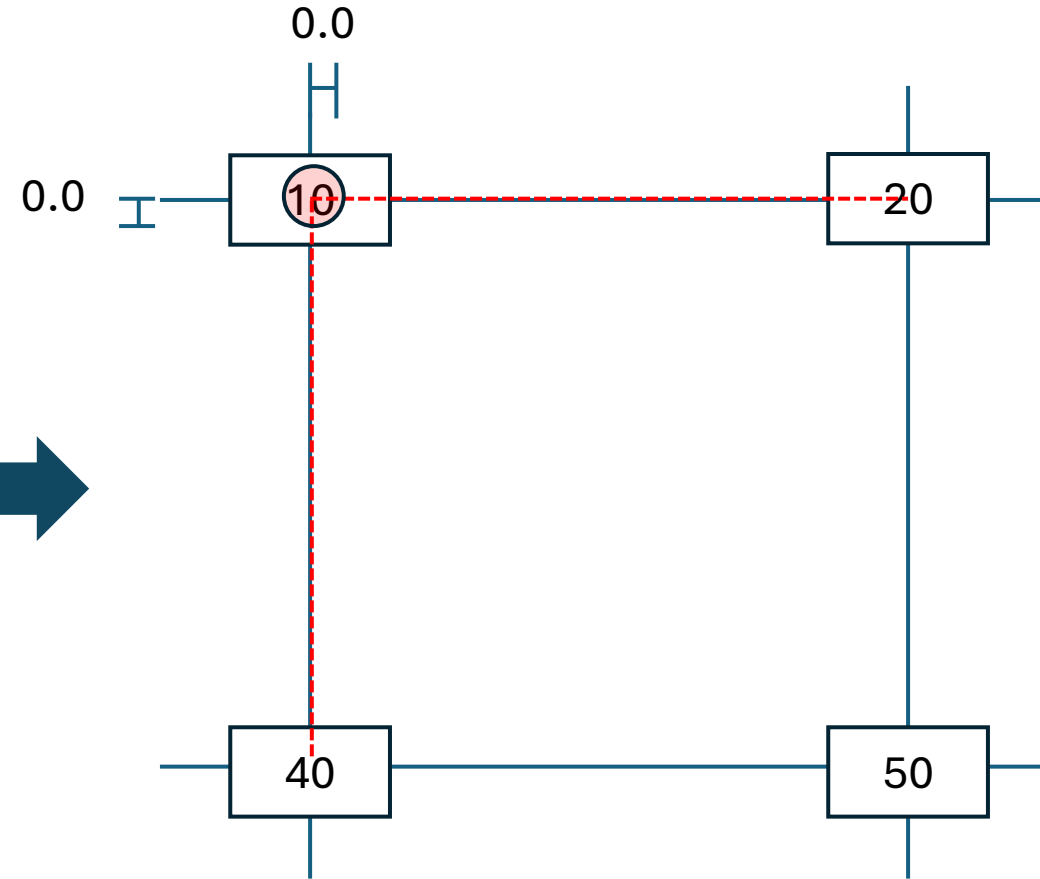
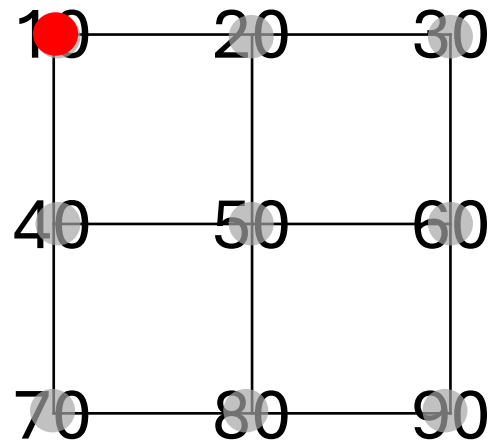
# Step3:計算內插值(3)



# Step3:計算内插値(4)

$$0.6667 \times 0 = 0$$
$$0.6667 \times 0 = 0$$

|    |    |    |
|----|----|----|
| 10 | 20 | 30 |
| 40 | 50 | 60 |
| 70 | 80 | 90 |

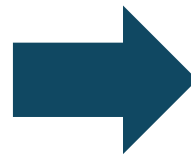


$$F(0,0) = 1 \times [0 \times 40 + 1 \times 10] +$$
$$0 \times [(0 \times 50 + 1 \times 20)] = 10$$



# 完成!

|    |    |    |
|----|----|----|
| 10 | 20 | 30 |
| 40 | 50 | 60 |
| 70 | 80 | 90 |



|    |      |      |    |
|----|------|------|----|
| 10 | 16.6 | 23.3 | 30 |
| 30 | 36.6 | 43.3 | 50 |
| 50 | 56.6 | 63.3 | 70 |
| 70 | 76.6 | 83.3 | 90 |

## 2-D 雙線性內插法 整理

### Step1:縮放比例 (Scaling Ratio)

- 原始圖像尺寸:寬度 $w_1$  , 高度 $h_1$
- 目標圖像尺寸:寬度 $w_2$  , 高度 $h_2$

$$S_x = \frac{w_1 - 1}{w_2 - 1} \quad S_y = \frac{h_1 - 1}{h_2 - 1}$$

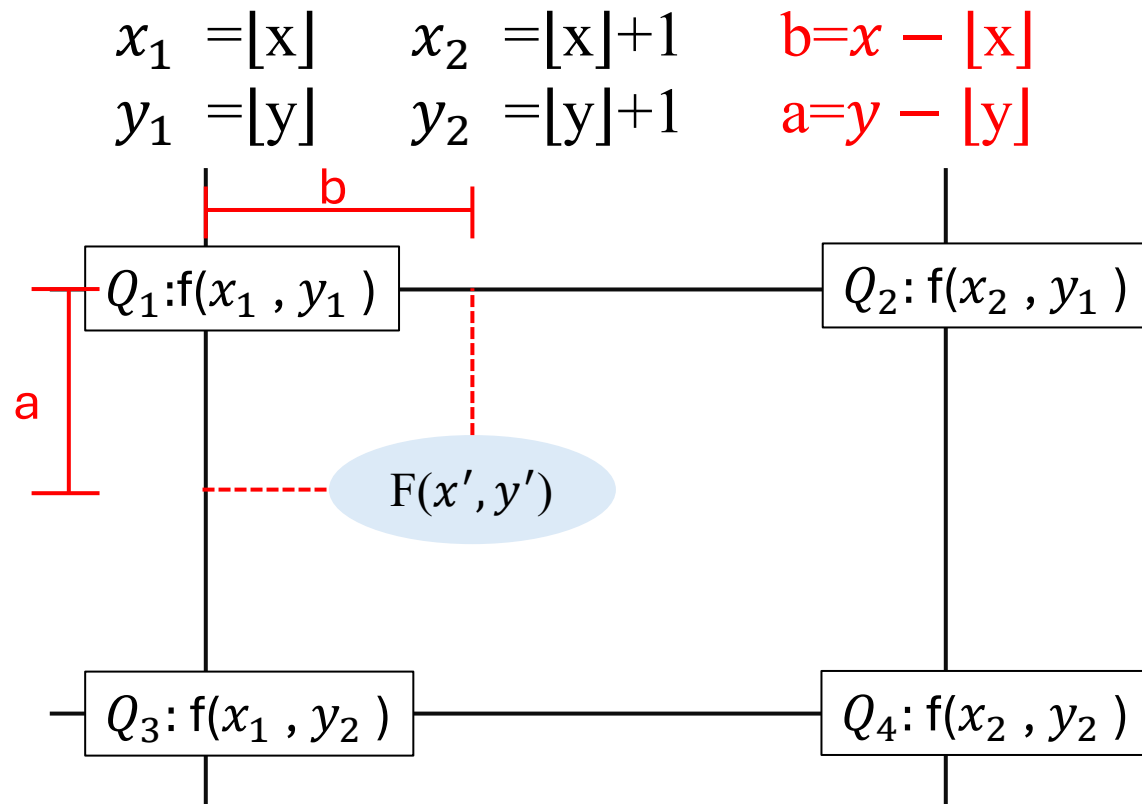
### Step2:映射到原圖

要算目標圖 $F(x', y')$ 的值  
先算映射到原圖的 $f(x, y)$

$$x = x' \times S_x$$

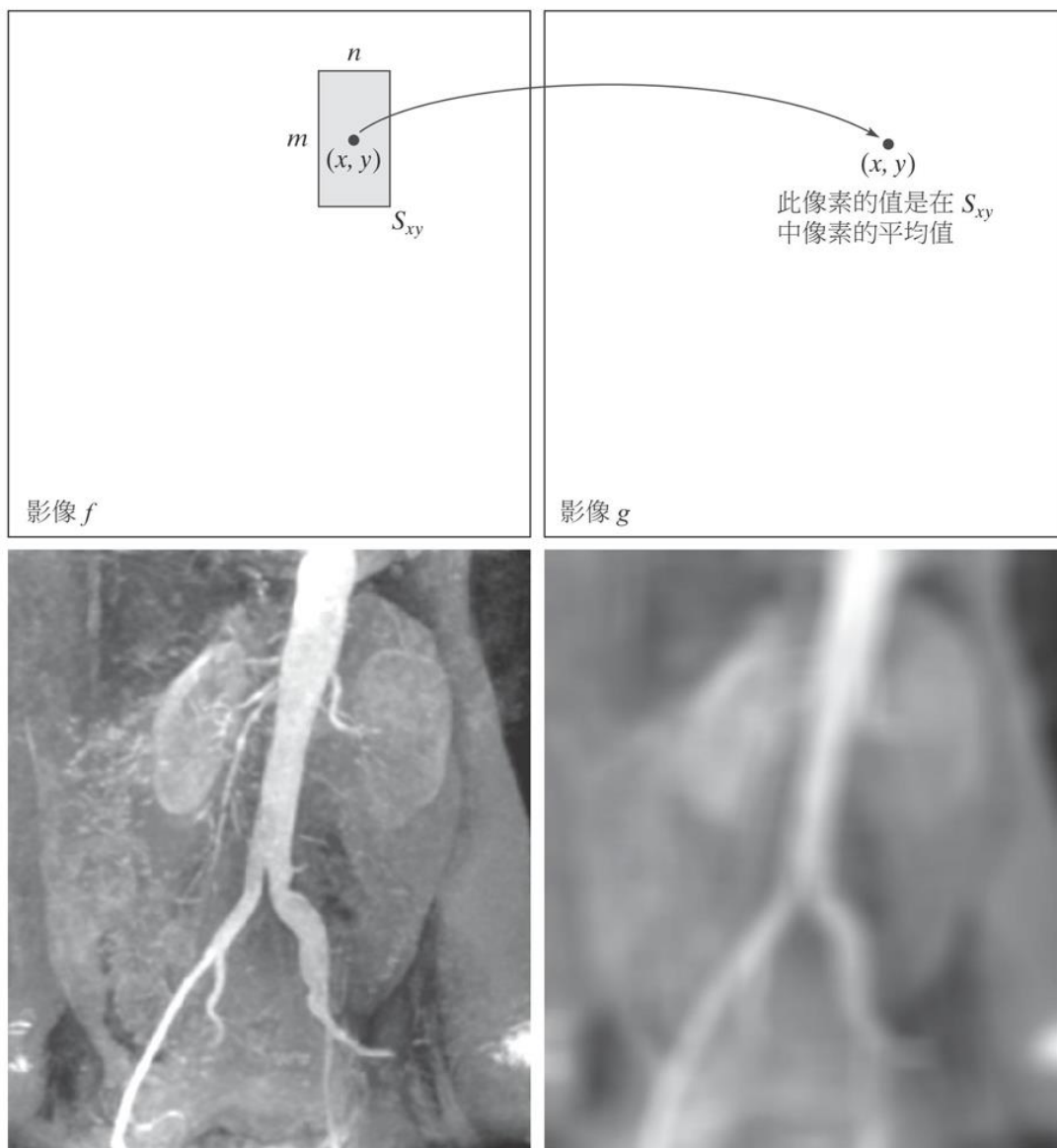
$$y = y' \times S_y$$

### Step3:計算目標圖值



$$F(x', y') = (1-b) \times [(1-a) \times Q_1 + a \times Q_3] + b \times [(1-a) \times Q_2 + a \times Q_4]$$

## 2.5 空間運算：鄰域運算



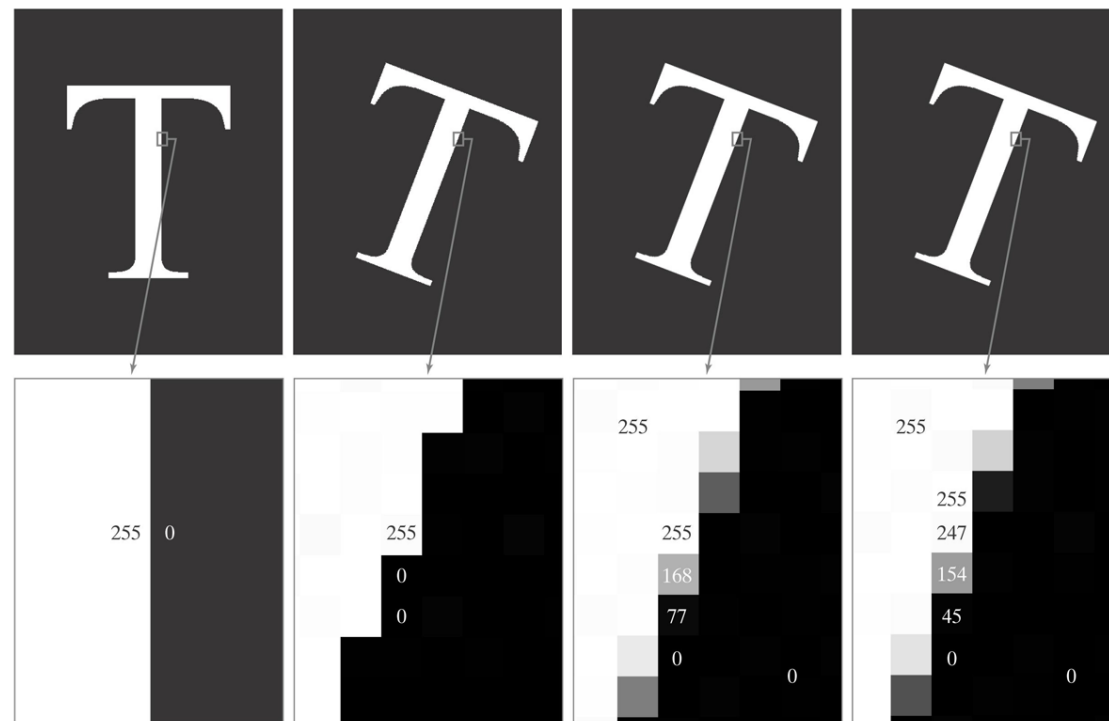
平均

$$g(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} f(r, c)$$

## 2.5 空間運算：仿射轉換

表 2.1 依據 (2-23) 式的仿射轉換

| 轉換名稱                                  | 仿射矩陣 A                                                                                                   | 座標方程式                                                                    | 例 子 |
|---------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----|
| 恆等式                                   | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                      | $x' = x$<br>$y' = y$                                                     |     |
| 尺度調整 / 翻轉 (對於翻轉，將一個尺度因子設成 -1 且其它設成 0) | $\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                  | $x' = c_x x$<br>$y' = c_y y$                                             |     |
| 旋轉 (對原點)                              | $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | $x' = x \cos\theta - y \sin\theta$<br>$y' = x \sin\theta + y \cos\theta$ |     |
| 平移                                    | $\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$                                  | $x' = x + t_x$<br>$y' = y + t_y$                                         |     |
| 切變 (垂直)                               | $\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                    | $x' = x + s_v y$<br>$y' = y$                                             |     |
| 切變 (水平)                               | $\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                    | $x' = x$<br>$y' = s_h x + y$                                             |     |



a b c d  
e f g h

圖 2.29 (a) 字母 T 的 541×421 影像；(b) 用最近鄰插補法做強度指定給經旋轉  $-21^\circ$  的影像；(c) 用雙線性插補法使影像旋轉  $-21^\circ$ ；(d) 用雙立方插補法使影像旋轉  $-21^\circ$ 。(e)~(h) 放大部分 (每個方形是一個像素，而所顯示的數字是強度值)。

$$[x \ y \ 1] = [v \ w \ 1] \mathbf{T} = [v \ w \ 1] \begin{bmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ t_{31} & t_{32} & 1 \end{bmatrix}$$

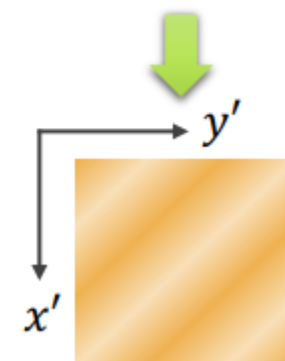
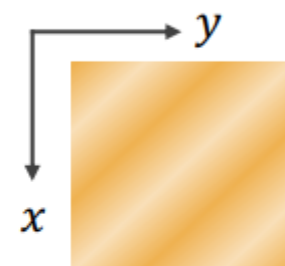
# (1)恒等式

- Identity transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Identity

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



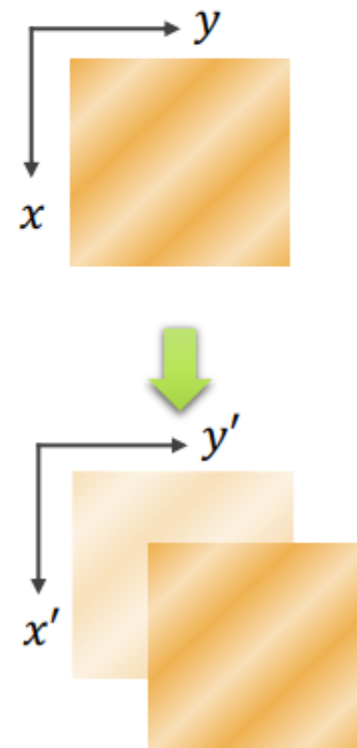
## (2) 平移

- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Translation

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



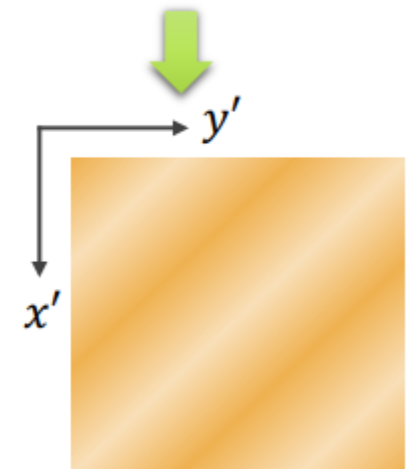
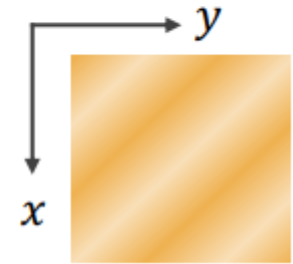
### (3) 放大縮小

- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Scaling

$$\mathbf{A} = \begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



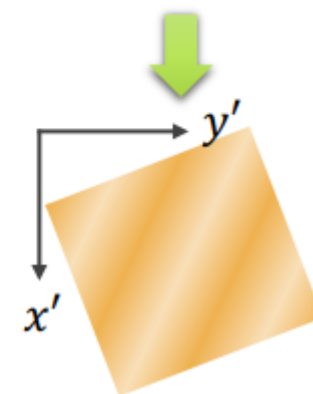
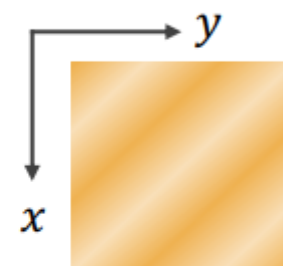
## (4) 旋轉

- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Rotation

$$\mathbf{A} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

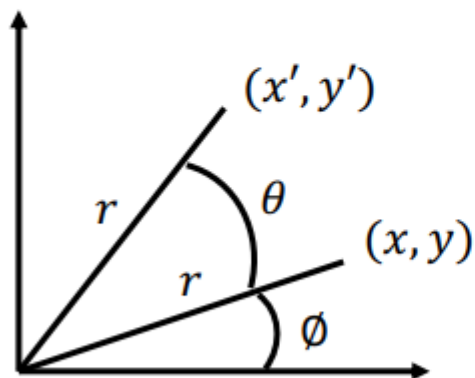




# 旋轉-推導過程

- Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



$$\begin{aligned} x &= r \cos \phi \\ y &= r \sin \phi \end{aligned}$$

$$\begin{aligned} x' &= r \cos(\phi + \theta) \\ y' &= r \sin(\phi + \theta) \end{aligned}$$

$$\begin{aligned} x' &= r \cos(\phi) \cos(\theta) - r \sin(\phi) \sin(\theta) \\ y' &= r \sin(\phi) \cos(\theta) + r \cos(\phi) \sin(\theta) \end{aligned}$$

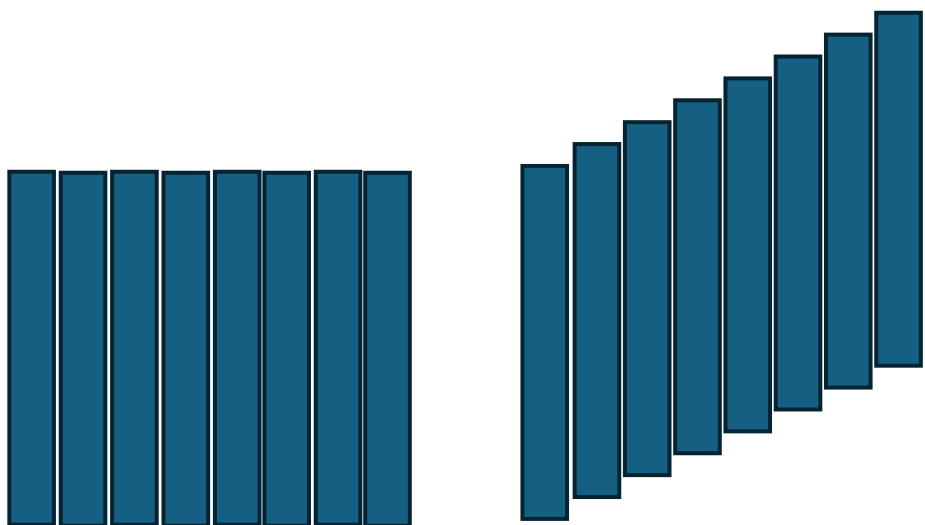
$$\begin{aligned} x' &= x \cos(\theta) - y \sin(\theta) \\ y' &= x \sin(\theta) + y \cos(\theta) \end{aligned}$$

## (5) 切變(垂直)

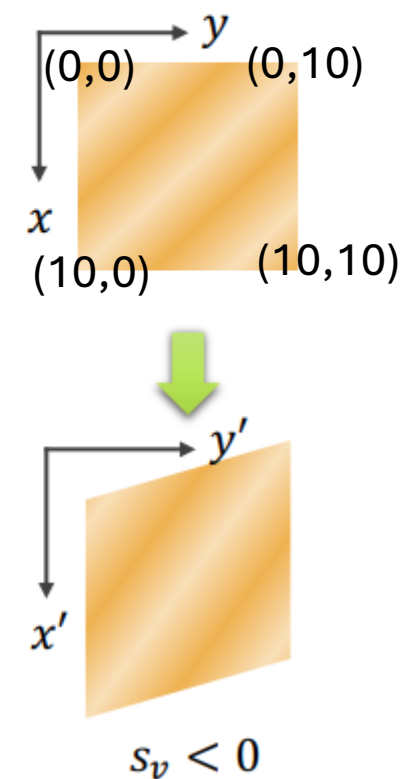
- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Shearing (vertical)



$$\mathbf{A} = \begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



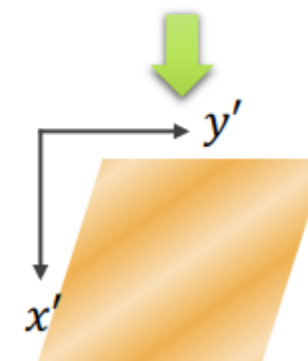
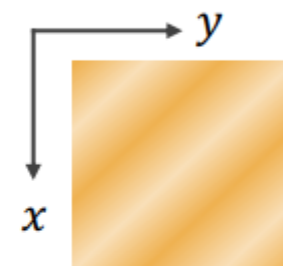
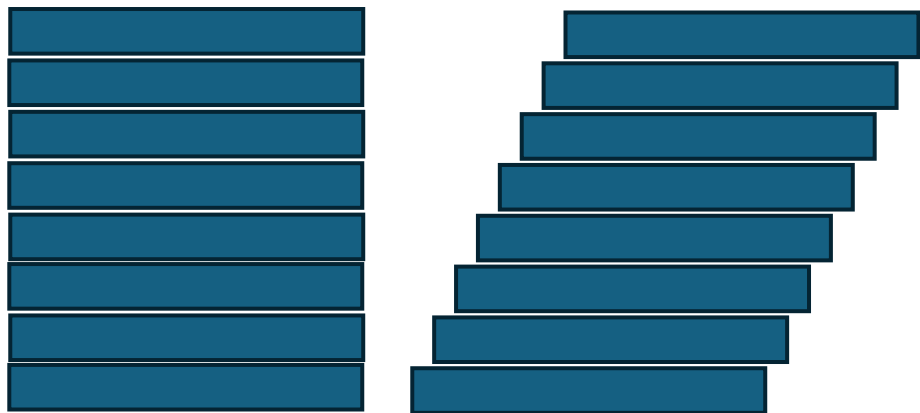
## (5) 切變(水平)

- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Shearing (horizontal)

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$s_h < 0$$

# 影像還原

- Forward mapping

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Backward mapping

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

# 小考會考

- 線性索引
- 雙線性內插
- 仿射轉換